

SoftwareX

BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research --Manuscript Draft--

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Abstract:	Occupants are a primary source of uncertainty in building energy consumption and management, yet existing occupant behavior models cannot capture adaptive and reasoning responses considering the occupant's personal history, current context, and the type of energy signal being delivered. This study presents BuildOcc, an open-source Python platform that grounds large language model agents in the American Time Use Survey (ATUS), a nationally representative diary dataset, using the 6,611 diary respondents who fall within four demographic strata. Through BuildOcc, each simulated occupant agent can be instantiated with a demographic persona drawn from ATUS population statistics, a memory stream that accumulates and reflects on timestep-level observations, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface (Python library, REST application programming interface (API), and Model Context Protocol server) so that any building energy tool (EnergyPlus, Home Assistant) can integrate behavioral intelligence without bespoke coupling code. A plugin registry lets the community add new occupant strata, custom schedulers, and alternative memory backends as separate installable packages. Two validation tiers show that the activity scheduler reproduces its ATUS reference distributions to within sampling noise and that demographic priors propagate into persona-consistent agent reasoning across timesteps, establishing internal consistency across strata. BuildOcc provides the building energy community with a reusable, openly available implementation of the occupant behavioral layer. BuildOcc is openly released at https://doi.org/10.5281/zenodo.22255938 under the Apache License 2.0 and installable via <code>pip install buildocc</code>.

Response to Reviewers — Second Revision

BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy
Research

Manuscript SOFTX-D-26-00798 — SoftwareX

Wooyoung Jung

September 4, 2026

Dear Editor and Reviewer,

The author is grateful to the Editor and to Reviewer 2 for a second round of unusually careful and constructive comments. Reviewer 2 examined the released software directly, recomputed the reported statistic against an independent null, and measured episode durations from the package’s own tables. The author appreciates the considerable effort this represents; it identified genuine defects, and both the software and the manuscript are more reliable as a result.

Before turning to the individual comments, three points apply throughout and are set out here to avoid repetition.

First, the reviewer inspected the archived v1.0.0 capsule, and every data claim in the review is correct for that artifact. The activity table shipped in v1.0.0 does have zero mass for hours 00:00–03:59 and does thin through the evening. The reviewer’s diagnosis is also correct: ATUS diaries run 04:00 to 04:00, and the pipeline was binning on the wall clock. The defect is corrected in v1.0.1.

Second, version 1.0.1 has been released, and a second defect is disclosed here that the review did not raise. While verifying the reviewer’s findings, the ATUS activity code map was found to be shifted by one subcategory group against the official ATUS lexicon. Three of the eight categories therefore measured the wrong activity: `food_prep` captured Housework (0201) instead of Food and Drink Preparation (0202), `laundry` captured Food presentation (020202) instead of Laundry (020102), and `tv` captured Relaxing (120301) instead of Television and movies (120303/120304). The reviewer could not have been expected to notice this, and it is reported here because it affects published numbers and the reviewer is entitled to a complete account. It is fixed in v1.0.1, and Table A.1 of the revised manuscript now states the corrected filters.

Third, every ATUS-derived number in the manuscript has been regenerated, including numbers the review did not question. Because the reported values were computed from the pre-fix tables, correcting the pipeline moved results for all four strata, not only for O4. Readers comparing the two versions will see changes throughout Tables 6, 7, 8 and C.1. The most consequential change is that the fixed baseline in Tier 1 falls from 9.92–11.48 to 1.07–2.64 nats: with the corrected reference, the epsilon-smoothing artifact that Reviewer 1 objected to in the previous round (the “1,224-fold” claim) no longer occurs, and the remaining gap is a genuine distributional difference. The Tier 2 experiments (Tables 7 and 8, Figure 4) were repeated on v1.0.1 because their activity inputs are drawn from the corrected tables; the agent code, prompts, model versions,

and seeds are unchanged. Price information (Type B) is now accepted in every stratum and direct commands (Type A) almost never, while social-norm signals remain unaccepted; Sections 3.3 and 4 have been rewritten to the new results. The Section 3.1 walkthrough was likewise re-run on v1.0.1, and its quoted reasoning is from that run.

The revised software is archived at [10.5281/zenodo.22255938](https://zenodo.org/record/22255938) (v1.0.1) and the corrections can be verified directly:

- every (`stratum`, `day_type`, `hour`) cell of `time_at_activity.csv` now sums to 100.0, for all 192 cells;
- `scripts/tier1_null.py` reproduces the sampling null reported in Table 6;
- `scripts/episode_durations.py` reproduces the duration comparison in Section 3.2;
- `scripts/measure_cost.py` reproduces the cost figures in Section 2.7;
- `scripts/validate_strata.py` and `scripts/validate_signals.py`, with the settings stated in Section 3.3, reproduce Tables 7 and 8; the result files are included in the repository.

Figures 2 and 3 were regenerated from the v1.0.1 result files. In doing so, a defect was found in the script that drew Figure 3 in the previous version: it overwrote the pooled “other” reference series instead of accumulating into it, which inflated the dotted reference curves; the shipped package was not affected. Finally, a small number of editorial changes were made that no comment requested: abbreviations are now expanded at first use, one abstract sentence was reworded, and bibliographic details were updated for references that have since been assigned volumes and page numbers.

In what follows, each comment is shown in *italics*, the response as bullet points, and the corresponding revised manuscript text in [blue](#). Cross-references are to sections and tables of the revised manuscript; exact line numbers will be supplied against the line-numbered proof.

Editor

Comment 1. *Check that the text (before the references) is <4000 words (and ≤ 6 figures) as specified in our guidelines. The paper looks like it is a bit long. You can use appendices and/or supplementary material as long as it follows the references.*

Response.

- Both limits are now met. The body text is **3,967 words**, reduced from 4,227 at the previous revision, and the manuscript contains **four figures** against a limit of six.
- Following the Editor’s suggestion, the three appendices now **follow the references**. In the tracked-changes file this appears as a large deletion in the old position and a matching insertion after the bibliography, an artifact of `latexdiff` having no representation for relocation; their content changed only where Reviewer 2’s comments required (Comment 1 and Minor 2–4).
- The reduction is net of material added at the reviewer’s request. The cuts removed duplication rather than substance: the per-timestep pipeline had been described three times (component list, architecture figure, and a five-step enumeration), and two results paragraphs recited numbers already tabulated beside them. This also completes the length reduction Reviewer 2 requested in the previous round.

Reviewer 2 — Main points

Comment 1. *The shipped activity table has no data for hours 00:00-03:59, and thins out through the evening. In `occupant_agent/data/time_at_activity.csv` the eight category percentages should total 100 in every (stratum, day type, hour) cell. They total 0.00 for hours 0-3, and 82 / 67 / 35 / 6 for hours 20 / 21 / 22 / 23. `scheduler.py` fills hours 0-3 by returning the literal “sleeping”. This looks like an ATUS diary-day boundary issue — diaries run 4 a.m. to 4 a.m. — so wrapping the day at 04:00 instead of binning on the wall clock should fix it in one place. Please check, and either fix it or state it as a limitation.*

Response.

- This is confirmed in full, and the reviewer’s diagnosis of the cause is exactly right. The observation that the fix belongs in one place proved correct as well: it is a single change to the time axis in `scripts/atus/analyze.py`.
- ATUS start and stop times are ordinary clock times in [00:00, 23:59]; the 04:00-to-04:00 diary ordering is carried by activity sequence, not by an extended 24–27 h hour code. The original code assumed the extended encoding, so every episode crossing midnight was dropped and hours 00:00–03:59 received no mass. Episodes are now mapped onto a 04:00-anchored axis before the hourly query.
- All 192 (stratum, day_type, hour) cells now sum to 100.0.
- The literal “sleeping” fallback has been removed. It existed only to paper over the empty cells and asserted a category the data did not support. The remaining fallback is uniform and is unreachable for the bundled tables.
- Every ATUS-derived value in the manuscript has been regenerated against the corrected tables.

Revised text (Appendix A):

ATUS diaries run from 04:00 to 04:00 the following day, and episode times are recorded as ordinary clock times. Episodes are therefore placed on a 04:00-anchored axis before the hourly query, so that an episode crossing midnight is credited to the diary day it belongs to.

Comment 2. *A cross-check inside the package. `time_of_day_distributions.csv` puts ~52% of O1 sleeping time as starting between 20:00 and 23:59, while the table above gives $P(\text{sleeping}) = 0.02\text{--}0.08$ across those same hours. Worth reconciling the two files.*

Response.

- Both figures are confirmed: the start-time file does place 51.8% of O1 sleeping mass in 20:00–23:59.
- The two files are not in conflict, because they answer different questions. `time_at_activity.csv` is a *time-in-activity* rate — the share of the population engaged in a category *at* a clock hour, measured by episode overlap at the half-hour mark. `time_of_day_distributions.csv` is a *start-time* distribution — how one category’s episode *onsets* are spread across the day.
- A category can therefore have most of its onsets in the evening while the probability of being in it at any given evening hour stays low, which is exactly the pattern for sleep.
- The reviewer’s difficulty in telling them apart was justified: the data-file documentation shipped inside the package (`occupant_agent/data/README.txt`) listed neither the distinction nor the

start-time file itself. It now defines both semantics explicitly, together with the diary axis and the weight variable.

- Only `time_at_activity.csv` is consumed by `ActivityScheduler`; the start-time file is diagnostic.

Comment 3. *Tier 1 needs a null to be interpretable. The scheduler draws from P and Tier 1 then compares P against draws from P , so the reported values partly reflect sampling noise. Drawing from the package’s own tables and evaluating Eq. (1) at the same n gives 0.014–0.030 nats — the same range as the reported 0.0092–0.0253. The fix is small: state n per (hour, day-type) cell and report the null alongside. Consider renaming the tier “sampler correctness”, which is what it shows, and which is a fair thing to claim.*

Response.

- The reviewer’s characterization is accurate, and the proposed rename has been adopted: Section 3.2 is now titled **Tier 1: Sampler Correctness**. The revised title describes what the tier measures, which the original did not.
- The null has been computed as suggested and is reported alongside each value in Table 6, with n stated: 180 simulated days place **520 weekday observations in each hourly cell**. The null is obtained by drawing 520 samples per hour directly from the reference and scoring them with Equation 1, over 1,000 repetitions.
- The result supports the reviewer’s reading. Every ATUS-grounded value falls inside or below its own null interval, so the scheduler is statistically indistinguishable from a correct sampler. The manuscript now claims exactly that and no more.
- It is also worth recording that this tier cannot speak to the language model at all: the statistic is computed from the scheduler’s sampled category, with no model call in the path.
- The reviewer’s own estimate (0.014–0.030 nats) was made against the uncorrected tables; on the corrected tables the null is 0.0070–0.0076 nats. The reported values move correspondingly.

Revised text (Section 3.2):

The scheduler samples from the same table it is scored against, so the benchmark for a correct sampler is not zero: at finite n the realized distribution departs from P by chance alone. Table 6 therefore also reports a null obtained by drawing 520 samples per hour directly from P and scoring them with Equation 1, over 1,000 repetitions. ... Every ATUS-grounded condition falls inside or below its sampling null, so the scheduler reproduces its reference tables to within sampling noise.

Comment 4. *Please add one duration statistic. `sample()` depends only on hour and day type, so episode lengths are geometric by construction and per-hour marginals cannot detect it. Measured against the durations the package itself ships in `activity-frequency_*.csv`: mean sleep episode 67 min vs 338 real, work 28 vs 203, and 29–41 transitions per day. Reporting mean episodes/day and the episode-length distribution would close this — both references are already in the package. If a duration model is out of scope, that is fine, but please say so in Section 2.2.*

Response.

- The reviewer’s measurements were reproduced independently before replying, and they are correct: against the shipped diary durations, mean simulated work episodes run 28 min versus

203 min for O1, matching the reported figures exactly.

- Both requested items have been added. Section 3.2 now reports the duration comparison across all four strata, and Section 2.2 states plainly that no duration model is implemented.
- The mechanism is stated rather than described obliquely: because each timestep is drawn independently from the hourly distribution, episode lengths are geometric by construction, and no set of per-hour marginals can reveal it.
- A duration model is out of scope for this release, as the reviewer allows.

Revised text (Section 3.2):

Per-hour marginals do not constrain episode length, however. Because each timestep is drawn independently, episode lengths are geometrically distributed by construction, each step ending the episode with a fixed probability: simulated sleep episodes average 52–71 min across strata against 314–338 min in the diaries, and agents make 49–54 activity transitions per simulated day.

Revised text (Section 2.2):

The scheduler draws each timestep independently from the hourly distribution, so it reproduces the hourly marginals without modeling how long an activity lasts, and episode lengths are consequently shorter than the diaries record (Section 3.2); no duration model is implemented in this release.

Comment 5. *Two citation fixes in Section 3.3. Gyamfi & Krumdieck (2011) is a stated-preference survey on price / environment / supply security; Albadi & El-Saadany (2008) is a power-systems review of program types. Neither contains a peer-comparison result. The claim itself is defensible for peak events — Ito, Ida & Tanaka (2018), AEJ: Econ. Policy 10(1), 240-267, 10.1257/pol.20160093, measured moral suasion decaying to insignificance by the third event. For billing-cycle conservation the evidence runs the other way (Allcott 2011, ~2%). Suggest swapping the citations and narrowing the sentence to peak events.*

Response.

- The reviewer is correct that neither cited work reports a peer-comparison result. Both suggested references have been added in their place, and the author is grateful for the specific citations.
- The sentence has been narrowed to peak events, and the billing-cycle evidence that runs the other way is now stated rather than omitted, since reporting only the supporting side would repeat the original error in a different form.
- The remaining occurrences of the same citation pair were audited. At two of them the pair supports the phrase “established behavioral intervention typologies”, which Albadi & El-Saadany does support, being a taxonomy of demand-response program types; those are retained. Only the peer-comparison claim was misattributed.

Revised text (Section 3.3):

Social norm signals (Type C) drew no acceptance in any stratum. Field evidence on the nearest interventions is split: appeals to conserve during repeated peak events lose their effect by the third event [Ito et al. 2018], whereas periodic reports comparing a household’s use with its neighbors’ sustain savings of about 2% [Allcott 2011]. Type C is a single peak-event message, so only the first finding bears on it.

Comment 6. *Please name the model version. The paper correctly says reproducible studies should record the provider and model version, but no version string appears for any reported result. One sentence in Section 3.*

Response.

- The observation that the manuscript recommended a practice it did not itself follow is a fair one, and the omission has been corrected.
- The versions are now stated in Section 3, as the reviewer suggested, and are recorded from instrumented runs rather than read off code defaults.
- The advice sentence that the reviewer quoted has been shortened, since retaining it verbatim while also following it duplicates the point.

Revised text (Section 3):

All LLM results in this section use Anthropic models at $T=0$ with a 512-token response cap: `claude-haiku-4-5-20251001` for per-timestep decisions and `claude-sonnet-4-6` for reflection synthesis; Tier 1 involves no model calls.

Comment 7. *Missing literature. No citation to the time-use-survey occupancy modelling lineage (Richardson 2008; Widén & Wäckelgård 2010; Wilke 2013; Flett & Kelly 2016; on the ATUS side Mitra 2020, Malekpour Koupaei 2022). Two reviews cover it in one citation each: Osman & Ouf (2021), Build. Environ. 196, 107785; Vosoughkhosravi et al. (2023), Energy Build. 294, 113245. Also Deng & Peng (2026), Buildings 16(5), 887 — the closest published LLM occupant-agent work. A first-order Markov chain on the same ATUS data would be a stronger baseline than the fixed schedule, and would be one BaseScheduler subclass — a neat demonstration of the plugin system.*

Response.

- The reviewer’s point is well taken. A manuscript that grounds itself in time-use data should engage with the body of occupancy modeling built on the same kind of data, and the original text did not. All nine suggested references have been added, and the passage has been rewritten to describe what those models do and where their limits lie.
- The four ATUS-side references are cited where they bear on the method rather than in the introduction: Osman & Ouf (2021) in Section 2.6, on harmonized diary conventions, and Mitra et al. (2020), Malekpour Koupaei et al. (2022), and Vosoughkhosravi et al. (2023) in Appendix C, where the selection of load-relevant activity categories is grounded in the ATUS microdata.
- Deng & Peng (2026) has been added to the discussion of recent LLM occupant agents.
- On the Markov baseline: the suggestion is a good one and is not adopted in this round. Adding a new comparison condition would require new runs and new results text in a manuscript already reduced to meet the word limit. It is instead identified as the natural next baseline, and the manuscript already notes that a Markov chain is one of the grounding sources substitutable through BaseScheduler. Flett & Kelly (2016), now cited, is itself a higher-order Markov occupancy method.

Revised text (Section 1):

However, most existing models rely on fixed schedules (e.g., ASHRAE 90.1 reference schedules), which assign the same profile to every day and every occupant. Stochastic models improved

on this by sampling from sensor data or national time-use diaries: early work used first-order Markov chains to generate occupant behaviors with realistic day-to-day variation [Richardson et al. 2008; Widén & Wäckelgård 2010]; later models added explicit behavior episodes with durations [Wilke et al. 2013] and higher-order transitions that distinguish occupant types [Flett & Kelly 2016]. Each step reproduces occupant behavior statistics more faithfully, but all remain statistical descriptions.

Reviewer 2 — Minor points

Minor 1. *Table 2 defines O4 as “Unemployed adult — Not employed, age 25-44”. In ATUS these are different filters ($TELFs \in \{3,4\}$ vs $\{3,4,5\}$), differing by roughly an order of magnitude, and $n = 107$ matches the narrower one. Worth confirming — if the intended stratum is not employed, the sample is much larger and the R1 noise caveat could be dropped.*

Response.

- The reviewer was right to ask. On investigation the discrepancy proved to be a coding error rather than a labeling choice. TELFS has five values; the code documented four and treated $TELFs=4$ as “not in labor force”, when 4 is “unemployed, looking” and 5 is “not in labor force”. O4 therefore selected unemployed respondents only, while both its label and the manuscript described it as not employed.
- A second variable was also misused. TERET1 was treated as a retirement flag; the 2023 ATUS Data Dictionary defines it as “do you currently want a job”, in universe only for $TELFs=5$ aged 50 or over. It cannot apply to a 25–44 stratum and the screen has been removed.
- Corrected, **O4 comprises 677 respondents** and matches its stated definition. The four strata now total 6,611.
- As the reviewer anticipated, the small-sample caveat has been dropped; the limit the manuscript now states for the Tier 2 pattern is the number of simulation seeds per cell.
- Membership of O1–O3 is unaffected: their predicates are disjoint from O4’s, and their rows in the regenerated tables are byte-identical.

Revised text (Table 2):

O4 — Not-employed adult — Not employed (unemployed or not in the labor force), age 25–44.

Minor 2. *Table C.1: the caption restricts the work-location columns to employed respondents with a work episode, but values are given for the two non-employed strata; and the columns sum to 94.5 / 58.2 / 94.6 / 98.4 rather than 100. Publishing the exact variable filters would settle both.*

Response.

- Both observations are correct and have a single cause: the caption described a restriction to employed respondents that the computation does not apply, and the residual TEWHERE mass — locations that are neither home nor workplace — was omitted.
- That missing mass is exactly the shortfall the reviewer computed. A “% elsewhere” column has been added and **every row now sums to 100.0**.
- The caption now states the actual computation: the split of each stratum’s work time (ATUS codes 05, weighted minutes, episodes with a valid TEWHERE code) across home, workplace, and

elsewhere, computed for every stratum, with a note that the O2 and O4 values rest on the small amount of work time those strata report.

Revised text (Table C.1 caption):

Location columns split each stratum’s work time (ATUS activity codes 05, minutes weighted by TUFINLWGT, episodes with a valid location code) across the respondent’s home, a workplace, and elsewhere; the three sum to 100%. For O2 and O4 the split rests on the small amount of work time those strata report. The at-home share is the work-from-home probability of Section 2.

Minor 3. *Table A.1 lists eight categories; Figure 3 plots five — say which and why.*

Response.

- The figure caption now states exactly what is drawn: four load-relevant categories (sleeping, work, television, and eating) individually, with the remaining five pooled as “all other”. The figure has been regenerated from the corrected tables.
- Table A.1 now lists **nine** categories, travel having been separated from “other” (Minor 4), and its code filters have been corrected (see the disclosure on page 1).

Revised text (Figure 3 caption):

Hourly activity probability distributions for stratum O1 (employed single adult, weekday). Four load-relevant categories are drawn individually (sleeping, work, television, and eating) and the remaining five are pooled as “all other”. Curves are population probabilities over the weekdays of 180 simulated days (520 timesteps per hour, as in Table 6); an individual diary day is a step function, and the smoothness here is a property of that average. (a) ATUS-grounded simulation: bold lines track the dotted reference. (b) Fixed rule-based baseline: bold lines depart from the reference and misassign the O1 late-evening peak.

Minor 4. *“Other” holds 172 of the tier-3 codes, including travel; worth one line of acknowledgment.*

Response.

- Travel is now its own category rather than a component of the residual. The nine categories are sleeping, work, food preparation, laundry, television, eating, exercise, travel, and other.
- Table A.1 reports the tier-3 code count for each category against the corrected 300-code map: travel accounts for 31 codes and “other” for 201.

Revised text (Table A.1, rows added or changed):

Travel — prefix 18 — 31 codes — Travel related to work; to housework; to personal care.
Other — residual — 201 codes — Grooming; Reading; Computer use; Social communication.

Minor 5. *The abstract’s “16,684 respondents” — 6,041 are in the four strata; use that number.*

Response.

- The distinction matters, and the abstract and Section 2.2 now report the analytic sample instead of the size of the source survey.

- The figure is **6,611** rather than 6,041 because of the O4 correction in Minor 1; the reviewer’s arithmetic on the released version was correct.

Revised text (Abstract):

...using the 6,611 diary respondents who fall within four demographic strata.

Revised text (Section 2.2):

Of the 16,684 respondents surveyed, 6,611 fall within these four strata.

Minor 6. *Name the ATUS weight variable used. (2022 and 2023 both use TUFINLWGT, so pooling is fine.)*

Response.

- The variable is now named in Section 2.2, together with the reviewer’s own observation that it is common to both survey years and so permits pooling. It is also documented in the package’s data-file README (occupant_agent/data/README.txt), which previously omitted it.

Revised text (Section 2.2):

... then computes a weighted probability across respondents using the ATUS final person weight (TUFINLWGT). Weighting makes each probability an estimate for the U.S. population, and because the weight is defined identically in the 2022 and 2023 files the two survey years can be pooled.

Minor 7. *Section 2.7 says cost bounded the simulation length but gives no measured cost; one figure would help.*

Response.

- No measured figure could be given previously because the library discarded the token counts that every provider returns with each response. Token accounting has been added, and a measurement harness (scripts/measure_cost.py) now records usage from the provider’s own metadata and prices it from a dated table.
- Measured over two simulated agent-days for each of the four strata, one agent-day costs **\$0.19–\$0.21**.
- The measurement also corrected a claim in the same paragraph. The manuscript stated “up to 96 calls per simulated day”; the measured rate is 85.5–90.0, because activity-driven room transitions are resolved deterministically and never reach the model. The revised text says so.

Revised text (Section 2.7):

Measured over two simulated agent-days for each of the four strata (September 2026 list prices), one agent-day required 85.5–90.0 per-timestep calls and 1.5 reflection calls, consumed 111,000–122,000 prompt and 13,700–15,300 completion tokens, and cost \$0.19–\$0.21. Fewer than 96 calls are issued because room transitions that follow directly from the sampled activity, such as moving to the bedroom when sleep begins, are resolved without a model call; they are still counted as moves in Table 7. Wall-clock time was 2.1–4.3 s per timestep and is dominated by provider latency, not local computation.

Minor 8. *Figure 3’s caption contrasts “smooth transitions” with a step function; a real diary is a step function, and the smoothness comes from averaging over 180 days.*

Response.

- This is correct, and the original caption misattributed the smoothness to the model when it belongs to the averaging. The caption now states that an individual diary day is a step function and that the smoothness is a property of the 180-day average.
- The phrase “unrealistic step functions” has been removed for the same reason. The revised caption is reproduced under Minor 3.

Minor 9. *The Zenodo capsule predates this revision — deposit a new version if the code changed.*

Response.

- Done. Version 1.0.1 is released and archived at <https://doi.org/10.5281/zenodo.22255938>, and the concept DOI 10.5281/zenodo.21192894 resolves to the latest version.
- The deposit was verified after publication by downloading the archived capsule and checking it directly, rather than assuming the release tag propagated correctly.
- The manuscript, CITATION.cff and the repository README now cite v1.0.1, and the package published on PyPI (`pip install buildocc`) is also at 1.0.1, so the three distribution channels in Table 9 point to the same code.

Revised text (Table 1, rows C1 and C3):

C1 — Current code version — v1.0.1.

C3 — Permanent link to reproducible capsule — <https://doi.org/10.5281/zenodo.22255938>.

Minor 10. *The typeset PDF loses inter-word spaces in places (“ATUSmicrodata”) — a production issue.*

Response.

- The author’s own compiled output was checked for this and the defect does not reproduce: an automated scan of the typeset text found no lost inter-word spaces.
- The defect therefore appears to have been introduced during production rather than present in the submitted source. The publication office is asked to watch for it at proof stage, and the author will check the proof specifically for this.

The author thanks the Editor and Reviewer 2 once more for the care taken with this manuscript. The review identified two genuine defects in the released software and led to the discovery of a third; the corrected software and manuscript are both considerably more reliable as a result, and the author is grateful for the rigor that made this possible.

Sincerely,
Wooyoung Jung

BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research

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Abstract

Occupants are a primary source of uncertainty in building energy consumption and management, yet existing occupant behavior models cannot capture adaptive and reasoning responses considering the occupant's personal history, current context, and the type of energy signal being delivered. This study presents BUILD OCC, an open-source Python platform that grounds large language model agents in the American Time Use Survey (ATUS), a nationally representative diary dataset, using the 6,611 diary respondents who fall within four demographic strata. Through BuildOcc, each simulated occupant agent can be instantiated with a demographic persona drawn from ATUS population statistics, a memory stream that accumulates and reflects on timestep-level observations, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface—Python library, REST application programming interface (API), and Model Context Protocol server—so that any building energy tool (EnergyPlus, Home Assistant) can integrate behavioral intelligence without bespoke coupling code. A plugin registry lets the community add new occupant strata, custom schedulers, and alternative memory backends as separate installable packages. Two validation tiers show that the activity scheduler reproduces its ATUS reference distributions to within sampling noise and that demographic priors propagate into persona-consistent agent reasoning across timesteps, establishing internal consistency across strata. BUILD OCC provides the building energy community with a reusable, openly available implementation of the occupant behavioral layer. BUILD OCC is openly released at <https://doi.org/10.5281/zenodo.22255938> under the Apache License 2.0 and installable via `pip install buildocc`.

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Keywords: Artificial Intelligence Agent, Occupant Behavior Modeling, Building Energy Use and Management, American Time Use Survey

Required Metadata

Current code version

Table 1: Code metadata (mandatory)

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.1
C2	Permanent link to code/repository	https://github.com/humanbuildingsynergy/BuildOcc
C3	Permanent link to reproducible capsule	https://doi.org/10.5281/zenodo.22255938
C4	Legal code license	Apache License 2.0
C5	Code versioning system	Git
C6	Software code languages, tools, and services used	Python 3.11+
C7	Compilation requirements, operating environments & dependencies	<code>pip install buildocc</code> ; requires a large language model (LLM) API key (Anthropic, OpenAI, Google), or local Ollama
C8	Developer documentation	https://github.com/humanbuildingsynergy/BuildOcc/blob/main/README.md
C9	Support email for questions	wooyoungjung@arizona.edu

1. Motivation and Significance

Humans are complex systems that respond to environmental, social, and economic signals in ways that are difficult to predict, introducing substantial uncertainty in building energy use and management [1, 2, 3]. This long-standing challenge has motivated the development of occupant behavior models that can capture the unpredictable and adaptive nature of human decision-making [4]. However, most existing models rely on fixed schedules

(e.g., ASHRAE 90.1 reference schedules [5, 6]), which assign the same profile to every day and every occupant. Stochastic models improved on this by sampling from sensor data [7] or national time-use diaries: early work used first-order Markov chains to generate occupant behaviors with realistic day-to-day variation [8, 9]; later models added explicit behavior episodes with durations [10] and higher-order transitions that distinguish occupant types [11]. Each step reproduces occupant behavior statistics more faithfully, but all remain statistical descriptions.

Large language models (LLMs), with architectures for reasoning and memory, allow agents to simulate human-like decision-making through contextual reasoning, memory retention, and adaptive behavior—capabilities suited to modeling occupants in buildings. Recent studies demonstrated the potential of LLM-integrated agents to simulate occupant behavior in building energy contexts, showing that they can produce more realistic and adaptive responses to various energy signals compared to traditional models [4, 12, 13, 14, 15, 16]. Specifically, Nicholas et al. [17] proposed agents that adopt the generative agent memory architecture of Park et al. [18]—combining a persona-based core memory, a timestep-level memory stream, and periodic reflection—and demonstrate that this representation produces qualitatively different energy demand patterns than fixed-schedule or rule-based occupant models. Other agents extend this architecture in different directions: conditioning behavior on indoor environmental quality [19], negotiating a shared setpoint between occupants [20], responding to dynamic pricing [21], and adjusting setpoints under demand response [22]. However, these agents are instantiated from hand-crafted profiles or small convenience samples [23], few ground their schedules in national survey data, and few are released as open platforms—leaving no shared, survey-grounded starting point for reproducible occupant behavior research.

This study addresses that gap with BUILD OCC, an open-source platform for LLM-based occupant simulation. Personas and activity schedules are grounded in the American Time Use Survey (ATUS) [24], and each agent carries a memory stream that accumulates observations and periodically reflects on them, giving coherent behavior across a simulation day. A three-layer interface (Python library, REST API, and Model Context Protocol (MCP) server) connects the agent to EnergyPlus [25] co-simulation, Home Assistant, or a custom building management system, and a plugin registry lets other groups add strata, grounding sources, and memory architectures.

2. Software Description

Following an overview of the platform architecture, the platform is described in six parts: the ATUS behavioral grounding that drives activity scheduling and persona sampling; the inputs required at simulation initialization; the per-timestep LLM reasoning pipeline; the demand response signal interface; the extensibility mechanisms that allow the platform to be adapted to new research contexts; and the computational cost of running it.

2.1. Architecture Overview

BUILD OCC is organized around two levels of architecture: an internal *agent component architecture* that defines how occupant agents perceive, reason, and act within a simulation timestep, and a *platform interface* that exposes this functionality to external building energy tools (Figure 1).

2.1.1. Agent Component Architecture

The BUILD OCC occupant agent is composed of six interacting components:

- **Persona** — a structured demographic profile assigned to one of four strata (O1–O4; Table 2), with appliance ownership drawn from Residential Energy Consumption Survey (RECS) [26] priors, an income bracket drawn uniformly within the stratum’s RECS range, and a comfort band set from income.
- **ActivityScheduler** — samples activity at each timestep from stratum-, hour- and day-type-specific ATUS distributions (6,611 respondents; Appendix C).
- **MemoryStream** — an accumulating log of observations, actions, and reflections, ranked by recency and LLM-assigned importance following Park et al. [18] (Appendix B).
- **LLM Reasoning Engine** — queries a configurable provider (Anthropic, OpenAI, Google, or local Ollama) at temperature $T=0$, returning a structured action with a natural-language reasoning string.
- **SimulationEnvironment** — maintains device states, room occupancy, and thermostat setpoint between timesteps, exposing the current state before each decision and applying the returned action after it.
- **AgentStore** — persists the agent’s cognitive state (persona, memories, importance accumulator) to SQLite after each operation. Environment state is not stored: the calling program sends it with every request, so the server keeps nothing between calls.

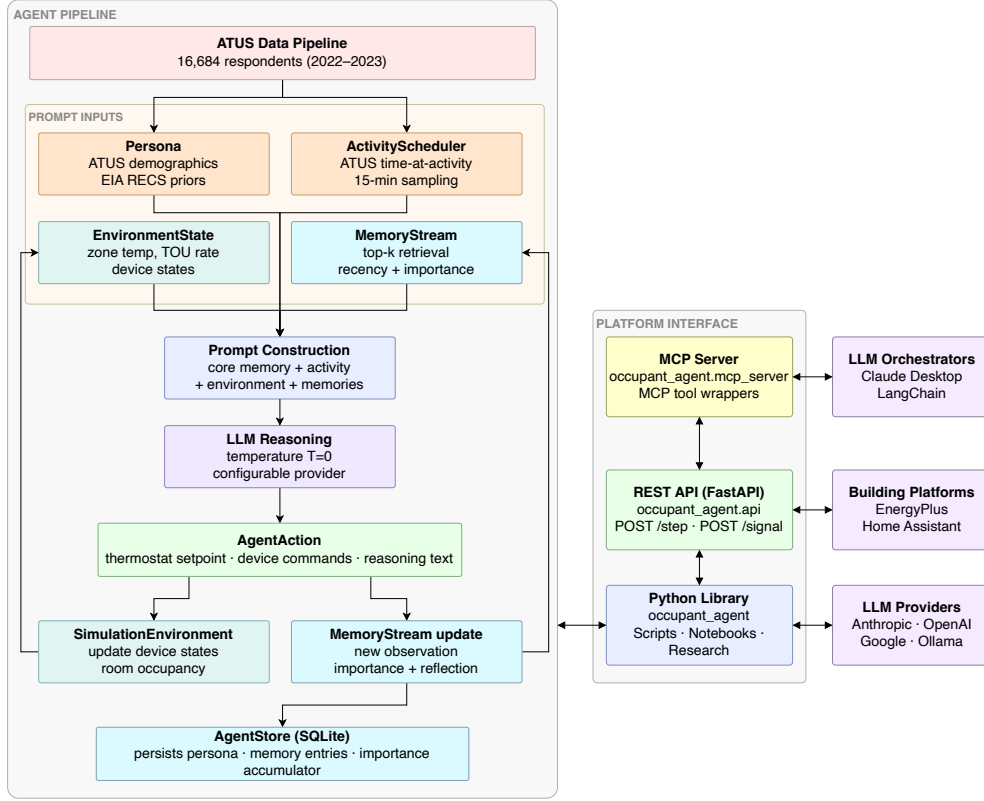


Figure 1: Architecture of BUILD OCC. *Left*: per-timestep agent pipeline showing how ATUS-grounded activity codes, retrieved memories, and environment state are assembled into an LLM prompt and how the resulting **AgentAction** updates both the simulation environment and the memory stream. *Right*: three-layer platform interface through which external building energy tools and LLM orchestration frameworks access the agent.

2.1.2. Platform Interface

BUILD OCC is distributed as the package `buildocc` and imported as `occupant_agent`. It exposes the agent through three integration layers, each targeting a different class of integrator:

1. **Python library** (`occupant_agent`) — direct access to all agent components; the recommended path for batch simulations.
2. **REST API** (`occupant_agent.api`) — a web server (FastAPI [27]) that keeps no state between requests, exposing agent initialization, per-timestep decisions (`POST /step`), signal delivery (`POST /signal`), and state query (`GET /state`); suitable for EnergyPlus co-simulation callbacks and any platform that can issue ordinary web (HTTP) requests.

3. **MCP server** (`occupant_agent.mcp_server`) — wraps the REST API as Model Context Protocol [28] tools, so any MCP-compatible host (Claude Desktop, LangChain, Home Assistant) can drive a simulation without additional integration code.

2.2. ATUS Behavioral Grounding

Activity schedules are derived from ATUS 2022–2023 microdata. Time-at-activity probabilities are aggregated across nine categories (sleeping, work, food preparation, laundry, television viewing, eating, exercise, travel, and other) by hour of day, stratified by demographic group and day type (week-day / weekend). The method follows the “what is the person doing at this moment?” framing: for each diary episode $[t_{\text{start}}, t_{\text{stop}}]$, the method queries whether the episode covers the half-hour mark of each hour h , then computes a weighted probability across respondents using the ATUS final person weight (TUFINLWGT). Weighting makes each probability an estimate for the U.S. population, and because the weight is defined identically in the 2022 and 2023 files the two survey years can be pooled. The formal computation is given in Appendix A. The scheduler draws each timestep independently from the hourly distribution, so it reproduces the hourly marginals without modeling how long an activity lasts, and episode lengths are consequently shorter than the diaries record (Section 3.2); no duration model is implemented in this release.

Four demographic strata are defined (Table 2), selected to span the variation in building occupancy and demand response participation that is most relevant to residential energy research (Appendix C). Of the 16,684 respondents surveyed, 6,611 fall within these four strata.

Table 2: Demographic strata

ID	Label	ATUS filter
O1	Employed single adult	Full-time employed, lives alone, age 25–44
O2	Retired couple	Age 65+, not employed, spouse or partner present
O3	Employed parent	Full-time employed, children present, age 35–54
O4	Not-employed adult	Not employed (unemployed or not in the labor force), age 25–44

Work-from-home (WFH) behavior is handled separately: a per-agent WFH probability encodes the fraction of working days spent at home (the share of each stratum’s work time that ATUS respondents report at home; Table C.1),

and a per-day Bernoulli draw determines WFH status for each simulated day. The WFH flag is included in the agent’s per-timestep prompt so that the LLM reflects both the ATUS activity code and the work location, preventing the scheduling model from assigning “work” activities that place the agent outside the home on days it is working remotely.

Each stratum uses stratum-specific appliance ownership priors from the U.S. Energy Information Administration’s RECS [26, 29] and a comfort band (in °C) that widens for lower income brackets, reflecting the documented tendency of lower-income households to accept wider thermal deviations to reduce heating, ventilation, and air-conditioning (HVAC) costs [30, 26, 31, 32].

2.3. User-Provided Inputs

Table 3 lists all inputs required at initialization; the only per-timestep value the calling application must supply is zone temperature, since it depends on the building’s physical response to occupant actions. To lower the barrier to first use, the library ships preset helpers for a representative household device inventory and room layout. A command-line interface offers the same presets and a plain-text (YAML) configuration file for users who prefer not to write Python.

Table 3: User-provided inputs to BUILD OCC. All are specified once at initialization, except zone temperature, which the calling platform supplies at each timestep.

Input	Type	Description
Stratum	String	Demographic profile: O1–O4
Seed	Integer	Controls persona attribute sampling, activity scheduling, and work-from-home day draws; the same seed produces an identical agent across runs.
LLM provider	String	Anthropic, OpenAI, Google, or Ollama
Device inventory	List	Appliance IDs, initial on/off states, rated power (W)
Room list	List	Rooms the occupant can occupy
Thermostat setpoint	Float	Initial comfort setpoint (°C)
Zone temperature	Source	Fixed value, CSV file, or live building platform
Outdoor temperature	Source	Fixed value, CSV file, EnergyPlus weather (EPW) file, or callable
Time-of-use (TOU) rate schedule	Source	Fixed value, CSV file, or tariff function

Without a live link, zone temperature can be a constant or a pre-computed comma-separated values (CSV) file; a one-week EnergyPlus sample is included.

2.4. LLM Reasoning Pipeline

At each 15-minute timestep the agent resolves its current ATUS code to a natural-language activity and an occupancy flag (a sleeping or laundry activity moves the agent to the matching room by rule, without a model call), retrieves up to five memories ranked by recency and importance (Appendix B), and assembles these with the persona and the environment state into a system and user prompt. The model returns one of four action types (Table 4) as a structured-text (JSON) object carrying a target, a value, and a one-sentence memory note, which is appended to the stream with an LLM-assigned importance score. When the cumulative importance exceeds a threshold (default 100), a reflection pass distills recent observations into three higher-order insights. Figure 1 shows the same sequence.

Table 4: The four action types returned by `step()` at each timestep, with their decision trigger and a representative example. Demand response is handled separately via `receive_signal()` (Section 2.5).

Action type	Trigger	Example
<code>do_nothing</code>	No environmental or activity condition warrants intervention	Agent is sleeping; no device is running
<code>adjust_thermostat</code>	Zone temperature leaves the comfort band, or TOU rate context	Raises setpoint during peak window to reduce HVAC cost
<code>toggle_device</code>	Activity–appliance alignment (on) or inactivity (off)	Turns off lights when leaving for work
<code>move_room</code>	Activity context suggests transitioning to a different room	Moves to bedroom at 23:00 for sleeping

The engine samples at $T=0$ by default, so the model returns its single most probable output and a fixed provider, model version, and prompt reproduce the same action on every run. That determinism is relative to one set of model weights: it does not hold across providers or across successive versions of a hosted model. Studies that must stay reproducible over time can run a fixed, locally hosted open model, whose weights do not change.

2.5. Demand Response Signal Interface

Alongside the per-timestep pipeline, BUILD OCC provides a dedicated signal delivery interface that can be invoked at any point in a simulation. Three signal types are supported (Table 5), following established behavioral intervention typologies [30, 33, 34]. Signals are delivered by passing a signal type, a free-text message, and the current environment state to the agent, which returns a response code (*accepted*, *rejected*, or *deferred*) alongside a natural-language reasoning string drawn from the agent’s memory context. Keeping the three types mutually exclusive enables factorial experimental designs in which signal type and demographic stratum are the primary independent variables.

Table 5: Demand response signal types supported by BUILD OCC, classified by behavioral lever and illustrated with representative example messages. All three types share the same delivery mechanism and response codes (*accepted*, *rejected*, *deferred*).

Type	Behavioral lever	Example message
A — Direct command	Regulatory compliance	“Turn off your HVAC for 30 min to support grid stability during today’s peak demand event.”
B — Educational	Economic information	“Your HVAC costs 3 times more per kWh at the current peak rate — raising the setpoint 1°C saves an estimated \$0.35 today.”
C — Social norm	Peer comparison	“75% of similar households in your area have reduced HVAC use during today’s peak event.”

2.6. Platform Extensibility

BUILD OCC is designed for adaptation across research contexts without modifying the core library. Extensibility is organized along three independent axes:

- **Occupant population** (`BasePersona`): new demographic profiles — commercial occupants, hotel guests, non-US populations, or any custom stratum — are registered by implementing `BasePersona`.
- **Activity grounding** (`BaseScheduler`): the built-in scheduler uses ATUS microdata; any other grounding source (a Markov chain, a physics-based occupancy model, or alternative activity data) can be substituted by implementing `BaseScheduler`.

- **Memory architecture** (`BaseMemoryStream`): the built-in recency-plus-importance retrieval can be replaced with retrieval by semantic similarity (embeddings) or any other scheme by subclassing `BaseMemoryStream`.

Extensions declare themselves in their own package configuration (Listing 1) and are found automatically when the library loads, so a third party can add a stratum or scheduler without modifying BUILD OCC’s own code:

Listing 1: `pyproject.toml` entry-point declarations for a third-party stratum and scheduler.

```

190 1 [project.entry-points."occupant_agent.strata"]
17    2 P5 = "my_package.personas:LowIncomeElderlyAlone"
18    3
19    4 [project.entry-points."occupant_agent.schedulers"]
20    5 homer = "my_package.schedulers:HomerScheduler"
21

```

Transfer to another country is a data-substitution task rather than a redesign: the scheduler consumes a table of activity probabilities by stratum, hour, and day type, so any national time-use survey following harmonized diary conventions [35]—the Multinational Time Use Study [36] or the Harmonised European Time Use Surveys (HETUS) [37], for example—can be substituted by regenerating that table behind `BaseScheduler`. What does not transfer automatically is the persona layer, whose appliance-ownership and income priors come from a US survey (RECS) and would need a national equivalent. Additional context—comfort indices [38], language-based comfort and heat-strain signals [39, 40], or air quality readings—can be injected by extending the agent class or by passing free text at each timestep, through either interface. The environment and signal schemas are fixed for reproducibility, and a testing module lets extension authors check, without an API key, that a new stratum or scheduler meets the platform’s contract.

2.7. Computational Cost and Scalability

Because the agent queries a language model at most timesteps, cost and runtime scale linearly with simulated agent-days. A 15-minute timestep gives at most 96 calls per simulated day per agent, plus occasional reflection calls, with responses capped at 512 tokens. Measured over two simulated agent-days for each of the four strata (September 2026 list prices), one agent-day required 85.5–90.0 per-timestep calls and 1.5 reflection calls, consumed 111,000–122,000 prompt and 13,700–15,300 completion tokens, and cost \$0.19–\$0.21. Fewer than 96 calls are issued because room transitions that follow directly from the sampled activity, such as moving to the bedroom when sleep begins, are resolved without a model call; they are still counted

as moves in Table 7. Wall-clock time was 2.1–4.3 s per timestep and is dominated by provider latency, not local computation. Two mechanisms reduce this cost. Any local open-weight model served through Ollama removes per-call cost entirely, at the price of reduced reasoning quality, and scheduler-only studies—including the Tier 1 validation reported in Section 3.2—require no model calls at all.

These figures are per agent. Agents are independent—each carries its own persona, memory, and schedule, and the REST API is stateless—so a multi-occupant household runs one agent per occupant, with call volume scaling linearly and the agents running in parallel.

3. Illustrative Examples

This section demonstrates the platform in three parts: a walkthrough of a single agent through a demand response event, followed by two quantitative tiers. Tier 1 validates the *ActivityScheduler*, testing whether ATUS-grounded sampling produces category distributions matching empirical ATUS reference data relative to a deterministic rule-based baseline; Tier 2 validates the *LLM agent decisions*, testing whether each stratum’s demographic priors propagate into distinguishable behavioral patterns under identical environment conditions. Both are necessary because a well-grounded scheduler and a well-calibrated reasoning agent are independent failure modes — one can be correct while the other is not. All LLM results in this section use Anthropic models at $T=0$ with a 512-token response cap: `claude-haiku-4-5-20251001` for per-timestep decisions and `claude-sonnet-4-6` for reflection synthesis; Tier 1 involves no model calls.

3.1. Demand Response Walkthrough

To illustrate agent behavior during a residential demand response event, this subsection traces an O1 agent (28-year-old employed male, income bracket 9 on the 16-level RECS scale, comfort band $\pm 1.1^\circ\text{C}$ around the setpoint) through the peak electricity period of a summer weekday evening (18:00–19:45, TOU rate \$0.22/kWh). Zone temperatures come from pre-computed EnergyPlus output included in the repository [41]; outdoor temperature follows a synthetic summer-day profile. The simulation loop is:

Listing 2: Running the BuildOcc simulation loop with the sample zone temperature CSV.

```
1 # Full imports available in examples/simulation_loop.py (
2 repository)
255 from datetime import datetime, timedelta
3
4 agent = OccupantAgent.from_stratum("O1", seed=42)
```

```

5 scheduler = ActivityScheduler(stratum="01", seed=42)
6 zone_temp = zone_temp_from_csv(...) # EnergyPlus zone
7     temperatures
8 price_message = ("Your HVAC is running during peak hours
9     (4-9pm) at $0.22/kWh -
10     "three times the off-peak rate. Raising
11     the setpoint by 1°C saves ~$0.35
12     today.")
13
14
15
16 sim = SimulationEnvironment(
17     initial_devices=persona_devices(agent.persona),
18     initial_rooms=[RoomState(room_id="living_room",
19         occupied=True)],
20     thermostat_setpoint=22.0,
21     outdoor_temp_fn=summer_day_temp,
22     tou_rate_fn=peak_tou_rate,
23 )
24
25
26
27 base = datetime(2025, 8, 11, 18, 0)
28 for i in range(8):
29     ts = base + timedelta(minutes=15 * i)
30     env = sim.observe(ts, zone_temp(ts))
31     if i == 2:
32         agent.receive_signal("B", price_message, env)
33         action = agent.step(env, atus_code=scheduler.sample(
34             ts))
35         sim.apply(action, ts)
36         print(ts.strftime("%H:%M"), action.action_type,
37             action.reasoning)

```

At 18:00 (Step 1), the zone is 23.9°C against a 22°C setpoint, with 35°C outdoors and the peak rate in force. The sampled activity is work-related travel, so the agent is on its way home and raises the setpoint to 23.5°C:

"I'm away from home during peak rate hours (\$0.22/kWh), so raising the setpoint to 23.5°C will reduce HVAC runtime and save money while I'm not there to notice the slight warmth. The current 1.9°C deviation is already outside my comfort band, so this adjustment is justified both for cost and to prevent further temperature drift."

At 18:15 the agent arrives and switches on the television. At 18:30 (Step 3) a Type B educational signal is delivered: *"Your HVAC is running during*

300 peak hours (4–9pm) at \$0.22/kWh—three times the off-peak rate. Raising the setpoint by 1° C saves ~\$0.35 today.” The agent declines, on the strength of its own prior action:

305 *“I’m comfortable right now at 23.9° C and just settled in to watch TV. Raising the setpoint by 1° C would push me toward discomfort (my limit is ±1.1° C from setpoint), and \$0.35 savings isn’t worth feeling warm and uncomfortable for the next few hours.”*

For the next hour the agent holds the setpoint, recalling at each step that it adjusted the thermostat a short time ago; at 19:30, when the sampled activity turns to sleep, it switches the television off. This cross-timestep coherence—recalling an earlier decision and judging a new request against it—is what a fixed schedule cannot represent. The zone temperature comes from the pre-computed file and does not respond to the setpoint change, so the walkthrough is open-loop; closing that loop is the role of the co-simulation interface (Section 2.1.2).

315 3.2. Tier 1: Sampler Correctness

The ATUS-grounded scheduler is compared with a deterministic rule-based baseline by Kullback–Leibler (KL) divergence, which measures how far the simulated distribution departs from the reference. For each stratum, 180 days of 96 timesteps (17,280 observations) are simulated and the hourly category distribution is scored against the ATUS reference using:

$$D_{\text{KL}}(P\|Q) = \sum_c P(c) \log \frac{P(c)}{Q(c) + \varepsilon} \quad (1)$$

where P is the ATUS reference distribution, Q is the simulated distribution, and $\varepsilon = 10^{-9}$ is a smoothing constant. Table 6 and Figure 2 report the mean per-hour KL divergence over the weekday distribution, which places 520 simulated timesteps in each hourly cell. The scheduler samples from the same table it is scored against, so the benchmark for a correct sampler is not zero: at finite n the realized distribution departs from P by chance alone. Table 6 therefore also reports a null obtained by drawing 520 samples per hour directly from P and scoring them with Equation 1, over 1,000 repetitions. Every ATUS-grounded condition falls inside or below its sampling null, so the scheduler reproduces its reference tables to within sampling noise. The fixed baseline, which assigns deterministic activities by hour-of-day rules (e.g., sleeping 23:00–06:59, work 09:00–16:59) without demographic conditioning, diverges two orders of magnitude more, most of all for the retired (O2) and not-employed (O4) strata, whose schedules depart furthest from the

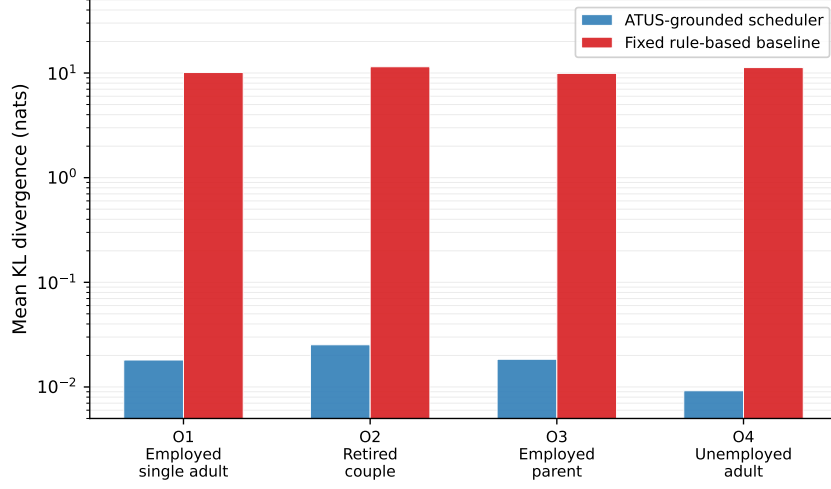


Figure 2: Mean per-hour KL divergence (nats, log scale) between simulated and ATUS empirical activity distributions over 180 simulated days, weekdays. The ATUS-grounded conditions sit at the sampling-noise floor for this sample size (Table 6).

Table 6: Tier 1 sampler correctness. Mean per-hour KL divergence (nats, natural-log units) against the ATUS reference, over 180 simulated days of 96 timesteps (520 weekday observations per hourly cell). The null is the divergence expected from a correct sampler at the same n , with its 95% confidence interval (CI), drawn directly from the reference distribution over 1,000 repetitions. Lower is better.

Stratum	ATUS scheduler	Sampling null (95% CI)	Fixed baseline
O1 (Employed single)	0.0077	0.0076 (0.0060–0.0091)	1.19
O2 (Retired couple)	0.0072	0.0070 (0.0056–0.0085)	2.64
O3 (Employed parent)	0.0073	0.0075 (0.0061–0.0091)	1.07
O4 (Not-employed)	0.0064	0.0070 (0.0057–0.0085)	2.13

employed-adult assumption. Figure 3 shows the contrast category by category for stratum O1 on weekdays: the ATUS-grounded simulation (panel a) closely tracks the reference distributions, while the baseline (panel b) collapses daytime activity into sharp step functions and misses the gradual evening wind-down.

Per-hour marginals do not constrain episode length, however. Because each timestep is drawn independently, episode lengths are geometrically distributed by construction, each step ending the episode with a fixed probability: simulated sleep episodes average 52–71 min across strata against 314–338 min in the diaries, and agents make 49–54 activity transitions per simulated day.

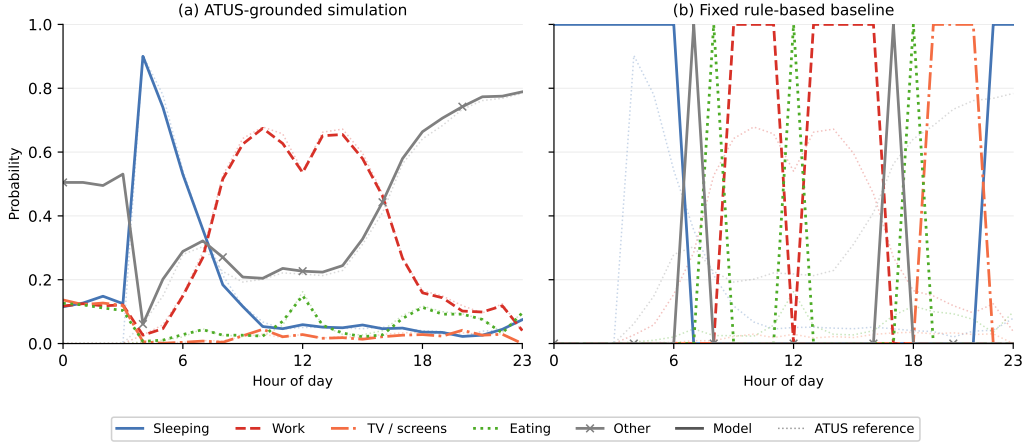


Figure 3: Hourly activity probability distributions for stratum O1 (employed single adult, weekday). Four load-relevant categories are drawn individually (sleeping, work, television, and eating) and the remaining five are pooled as “all other”. Curves are population probabilities over the weekdays of 180 simulated days (520 timesteps per hour, as in Table 6); an individual diary day is a step function, and the smoothness here is a property of that average. (a) ATUS-grounded simulation: bold lines track the dotted reference. (b) Fixed rule-based baseline: bold lines depart from the reference and misassign the O1 late-evening peak.

3.3. Tier 2: Persona-Driven Behavioral Differentiation

Tier 2 tests whether each stratum’s demographic priors produce distinguishable behavior. O1–O4 were each run for three seeds over one weekday (96 timesteps) under identical conditions—the same zone temperature, device configuration, and time-of-use tariff (\$0.22/kWh peak, \$0.08/kWh off-peak)—so that any difference is attributable to the persona.

Table 7 reports action-type distributions and mean thermostat setpoints during the TOU peak window (4–9 pm), and Figure 4 visualizes the active-action counts. The ordering follows daytime home occupancy: O2, at home all day, is the most active stratum, with the fewest inactive timesteps and the most device toggling and room movement, followed by O4; the two employed strata are the least active. Of the moves, 17–25 per stratum follow by rule from a sampled sleeping or laundry activity; the remaining 26–44 are model decisions. Every stratum ends the peak window with its setpoint 1.2–1.6°C above the 22°C it started at, O2 highest. Thermostat adjustments are rare in every stratum (1.7–2.4% of timesteps), so the strata differ in what they do with devices and rooms, not in how often they touch the thermostat.

To assess demand response participation, three standardized signals were delivered to each stratum (five seeds each) after a 10-timestep warm-up from

Table 7: Action-type distribution (%) per stratum: mean over three seeds, one weekday each (96 steps), giving 288 timesteps per stratum. At this sample size one percentage point is 2.9 timesteps, so differences of a few tenths should not be read as meaningful. Peak SP is the mean thermostat setpoint during TOU peak hours (4–9 pm, \$0.22/kWh).

Stratum	Inactive	Thermostat	Toggle	Move	Peak SP (°C)
O1 (Employed single)	75.3	2.4	6.2	16.0	23.2
O2 (Retired couple)	60.1	2.1	13.9	24.0	23.6
O3 (Employed parent)	77.1	2.1	4.9	16.0	23.4
O4 (Not-employed)	67.7	1.7	11.1	19.4	23.2

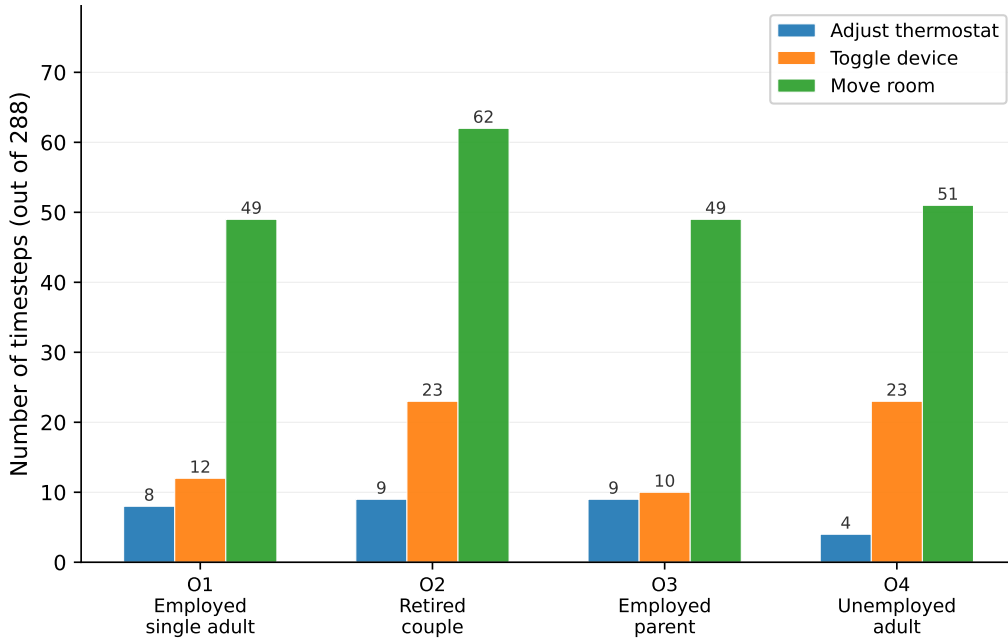


Figure 4: Active action counts per stratum across 288 timesteps (3 seeds $\times$ 96 steps, identical environment conditions). Do-nothing accounts for 60–77% of all timesteps across strata and is excluded to isolate cross-stratum differences in active decision-making. Bar labels show absolute counts.

16:00 at peak rates that seeds each agent’s memory: Type A (direct command: “Turn off your HVAC for 30 minutes”), Type B (educational price signal: a cost comparison stating the peak rate as three times off-peak), and Type C (social norm: “75% of similar households have reduced HVAC use”). Table 8 reports the number of accepting agents per stratum and signal type. Direct commands (Type A) were declined in all but one case: the agents reasoned that a 30-minute HVAC shutdown on a 35°C afternoon would carry

the zone past their comfort band. Educational price signals (Type B) were accepted in every stratum, from one agent in five (O3) to three in five (O4), and every acceptance cited the concrete dollar figure together with headroom inside the comfort band. Social norm signals (Type C) drew no acceptance in any stratum. Field evidence on the nearest interventions is split: appeals to conserve during repeated peak events lose their effect by the third event [42], whereas periodic reports comparing a household’s use with its neighbors’ sustain savings of about 2% [43]. Type C is a single peak-event message, so only the first finding bears on it.

Table 8: Signal compliance per stratum and signal type, reported as the number of accepting agents out of five random seeds. Percentages are not given, since five runs per cell cannot support them. Type A: direct command; Type B: educational price signal; Type C: social norm nudge. No agent deferred.

Stratum	Type A	Type B	Type C
O1 (Employed single)	0/5	2/5	0/5
O2 (Retired couple)	0/5	2/5	0/5
O3 (Employed parent)	0/5	1/5	0/5
O4 (Not-employed)	1/5	3/5	0/5

These results show that the ATUS demographic priors propagate into distinct LLM-driven behavioral patterns: daytime home occupancy drives device toggling and room movement, price information is the only signal accepted in every stratum, and neither a direct command nor a social comparison moves an agent that is already inside its comfort band. Three constraints bound the interpretation. First, Tier 2 establishes *internal consistency*: agents instantiated from different demographic priors behave differently, and in directions consistent with those priors. It does not establish behavioral realism, which would require comparison against measured occupant data and is left to future work [44]. Second, each stratum is run for one weekday per seed (three seeds for action distributions, five for signals), which shows direction but cannot establish significance; multi-day designs were constrained by API cost. Third, a single environment configuration is used, which limits generalizability across climate zones and occupancy contexts.

4. Impact

BUILD OCC lowers the barrier to four lines of research that are difficult to pursue with fixed-schedule occupant models:

Survey-grounded behavioral agents. BUILD OCC packages population-calibrated occupant agents as a reusable component rather than a per-project

1
2
3
4
5 infrastructure investment. Research groups can introduce demographically
6 differentiated occupant behavior into any building energy study without con-
7 structing a grounding pipeline from scratch. The ATUS-derived activity
8 distributions and RECS-conditioned personas provide a documented, survey-
9 grounded baseline that any researcher can adopt, extend, or cite as a reference
10 condition.

11
12 **Demographic behavioral experiments.** BUILD OCC’s combination of
13 ATUS-grounded demographic strata, persona income brackets, and config-
14 urable intervention signals enables controlled factorial studies of occupant
15 behavior. Unlike fixed-schedule models, each agent reasons from its de-
16 mographic persona and accumulated memory at every timestep, producing
17 adaptive responses that reflect personal history and current context rather
18 than predetermined patterns. Because both the activity schedule and the
19 agent’s reasoning are conditioned on the same demographic persona, re-
20 searchers can hold environment and intervention constant while varying stratum
21 — or vary the intervention while holding stratum constant — isolating
22 the contribution of socioeconomic and life-stage factors to any behavioral
23 outcome. The Tier 2 signal experiment (Table 8) illustrates this: educa-
24 tional price signals (Type B) were accepted in every stratum, from one agent
25 in five (O3) to three in five (O4), while direct commands were declined in all
26 but one case and social comparison in every case — differences in direction
27 that a fixed schedule cannot produce at all, having no mechanism to respond
28 to a signal. The signal types correspond to established intervention typolo-
29 gies in behavioral energy research [30, 33], so that results can be set beside
30 field studies that use the same typology.

31
32 **Platform-neutral integration.** The MCP server layer means that any tool
33 with MCP support—including Claude Desktop, LangChain, energy schedul-
34 ing systems [45], and custom orchestration frameworks—can drive the oc-
35 cupant agent inside a building simulation without writing integration code.
36 The REST API provides an equivalent path for EnergyPlus co-simulation
37 and Home Assistant integrations [12, 15].

38
39 **Community extensibility.** The plugin registry and abstract base classes
40 lower the barrier for researchers who want to add custom demographic strata,
41 alternative behavioral data sources or activity grounding pipelines, or novel
42 memory architectures. The conformance test suite enables rapid validation
43 of extensions before publication.

44 45 46 47 48 49 50 51 52 53 5. Conclusions

54 BUILD OCC delivers a survey-grounded, three-layer occupant agent platform
55 that the building energy community can use, extend, and build upon without
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reconstructing the behavioral layer for each study. Tier 1 confirms that the scheduler reproduces the ATUS activity distributions to within sampling noise; Tier 2 shows that demographic priors propagate into differentiated agent reasoning, though it does not by itself establish that the resulting behavior is realistic.

Beyond the implementation, the platform treats occupant behavior as a structured, demographically parameterizable experimental variable rather than an uncontrolled source of uncertainty. The ATUS-derived strata, frozen schema, and plugin registry are intended to make that variable reproducible and extendable across independent research groups.

Current limitations include a single-action constraint per timestep; ATUS-grounded strata covering US demographic profiles only; activity grounding restricted to primary activities, since ATUS records only one activity at a time; and a memory-importance score that the agent assigns to itself and then reuses for retrieval, without external calibration or a feedback path that would correct a mis-scored entry. Subsequent phases extend the platform with multi-agent household simulation, non-US occupancy data, and external behavioral benchmarks including ecobee Smart Thermostat field data [46] matched to the four ATUS strata.

CRediT Author Statement

Wooyoung Jung: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares that there are no competing interests.

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Current executable software version

Table 9: Software metadata (mandatory)

Nr.	Software metadata description	Please fill in this column
S1	Current software version	v1.0.1
S2	Permanent link to executables of this version	https://pypi.org/project/buildocc/
S3	Legal software license	Apache License 2.0
S4	Computing platforms / OS	Linux, macOS, Windows (Python 3.11+)
S5	Installation requirements & dependencies	<code>pip install buildocc</code> ; an LLM provider API key (Anthropic, OpenAI, Google) or a local Ollama model
S6	Link to user manual	https://github.com/humanbuildingsynergy/Build0cc/blob/main/README.md
S7	Support email for questions	wooyoungjung@arizona.edu

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A. ATUS Activity Probability Computation

This appendix provides the formal description of the activity probability computation summarized in Section 2.

Episode-overlap framing. ATUS diaries record each respondent’s day as a sequence of episodes, each with a start time t_s (minutes since midnight), stop time t_e , and a six-digit activity code in ATUS. For each hour h , the method identifies which episode was in progress at the half-hour mark:

$$c(r, h) = \text{activity code of respondent } r \text{ such that } t_s \leq h:30 < t_e \quad (\text{A.1})$$

where r indexes individual diary respondents. This in-progress query avoids the start-time bias that would overrepresent short activities (under a start-time count, a 5-minute meal and an 8-hour sleep would each count once).

ATUS diaries run from 04:00 to 04:00 the following day, and episode times are recorded as ordinary clock times. Episodes are therefore placed on a 04:00-anchored axis before the hourly query, so that an episode crossing midnight is credited to the diary day it belongs to.

650 *Category probability.* Nine activity categories are defined (Table A.1). The probability of category k at hour h , for demographic stratum s and day type $d \in \{\text{weekday}, \text{weekend}\}$, is:

Table A.1: The nine broad activity categories used in the two-step ATUS sampling procedure. ATUS activity codes are six-digit and hierarchical, so a filter is a leading prefix of the code; the third column counts the six-digit codes mapped to each category.

Category	ATUS code filter	Codes	Representative subcategories
Sleeping	prefix 0101	3	Sleeping; Sleeplessness
Work	prefix 05	14	Work (main job); Work (other job); Security procedures related to work
Food preparation	prefix 0202	4	Food and drink preparation; Food presentation; Kitchen and food clean-up
Laundry	code 020102 exactly	1	Laundry
Television	codes 120303, 120304	2	Television and movies; Television (religious)
Eating	prefix 1101	2	Eating and drinking
Exercise	prefix 13	42	Aerobics; Basketball; Biking; Swimming; Weightlifting
Travel	prefix 18	31	Travel related to work; to housework; to personal care
Other	residual	201	Grooming; Reading; Computer use; Social communication

$$P(k \mid s, d, h) = \frac{\sum_{r \in R(s, d)} w_r \cdot \mathbf{1}[c(r, h:30) \in k]}{\sum_{r \in R(s, d)} w_r} \quad (\text{A.2})$$

where $R(s, d)$ is the set of diary respondents matching stratum s on day type d ; w_r is the ATUS person-level survey weight (provided by the Bureau of Labor Statistics (BLS) to correct for sampling design); $\mathbf{1}[\cdot]$ is the indicator function that equals 1 when respondent r was engaged in a category- k activity at the half-hour mark $h:30$ of hour h ; and the denominator normalizes the weighted counts to probabilities. Evaluating Equation (A.2) for all $h \in \{0, \dots, 23\}$ and all nine categories yields a 24×9 weighted probability table per stratum per day type, which is bundled with the installed package.

Code-level sampling. Once a category k is sampled from $P(k \mid s, d, h)$, a specific activity code is drawn from within that category proportional to the total diary minutes logged across all stratum respondents for each code. This gives common activities realistic relative frequencies (e.g., sleeping in a bed is drawn far more often than napping on a sofa, even though both map to the “sleeping” category).

Work-from-home probability. Finally, the WFH probability p_{WFH} for each stratum is estimated from the ATUS `TEWHERE` variable — the “where were you when you did this activity?” field — restricted to employed respondents who worked on their diary day. p_{WFH} is the share of the stratum’s survey-weighted work minutes reported at home (the at-home column of Table C.1). A per-day Bernoulli draw from p_{WFH} then determines whether the simulated agent works from home on a given day.

B. Memory Retrieval and Reflection Mechanics

This appendix provides the mathematical detail for the memory retrieval and reflection logic implemented in `MemoryStream`, following the generative agent architecture of Park et al. [18].

Retrieval scoring. At each simulation timestep t , every stored memory entry m receives a combined retrieval score $s(m, t) \in [0, 1]$ that balances how recent the entry is against how important it was judged to be when written:

$$s(m, t) = 0.5 \tilde{r}(m, t) + 0.5 \tilde{i}(m) \quad (\text{B.1})$$

where $\tilde{r}(m, t)$ is the normalized recency score and $\tilde{i}(m)$ is the normalized importance score, both in $[0, 1]$ and weighted equally.

The recency component decays exponentially with a 24-hour half-life:

$$\tilde{r}(m, t) = 2^{-(t-t_m)/24\text{h}} \quad (\text{B.2})$$

where t_m is the simulation time (in hours) at which entry m was written, and $(t - t_m)$ is the elapsed time in hours since writing. The score equals 1.0 at the moment of creation and halves every 24 simulation hours, so memories from the previous day retain half their recency weight while week-old entries contribute near zero.

The importance component normalizes the integer score $i(m)$ assigned by the LLM at write time:

$$\tilde{i}(m) = \frac{i(m)}{10} \quad (\text{B.3})$$

where $i(m) \in \{1, \dots, 10\}$ is an integer on a scale from 1 (routine, nothing unusual) to 10 (rare, extreme situation), self-assigned by the agent’s LLM in the same JSON response that produces the behavioral action. The top- k entries ranked by $s(m, t)$ are injected into the next prompt ($k = 5$ by default).

695 *Reflection trigger.* An importance accumulator A tracks the cumulative importance of all entries written since the last reflection. After each entry m is stored, $A \leftarrow A + i(m)$. When $A \geq \theta$ (default $\theta = 100$), reflection is triggered: a dedicated LLM call receives the 30 most recent entries by simulation time and returns exactly three synthesized behavioral insights (e.g., “this occupant pre-cools before peak hours to avoid peak tariffs”). Each insight is written 700 back into the stream as a new entry with importance 9.0, and A resets to zero. At average importance 7–10 (high-salience events), reflection fires every 10–14 timesteps (≈ 2.5 – 3.5 simulation hours); at average importance 2–3 (routine observations), every 33–50 timesteps (≈ 8 – 12 simulation hours). In 705 the measured runs of Section 2, reflection fired 1.5 times per simulated day.

C. Demographic Strata Selection Rationale

The four strata were selected to maximize variation in the two behavioral dimensions most consequential for residential demand response research: *peak-hour occupancy schedule* and *household composition*. The selection is grounded 710 in the ATUS 2022–2023 microdata that drives the platform itself [47, 48, 49]. Table C.1 summarizes the occupancy metrics and sample sizes derived from ATUS 2022–2023 ($n = 16,684$ respondents). O1 and O3 are absent from the home for the majority of daytime hours (09:00–17:00), while O2 and O4 are home for most of the day: 60.9% and 62.6% of O1 and O3 weekday 715 hours in that window are spent in work activity, against 0.4% and 2.4% for O2 and O4. Within each occupancy cluster, household composition is varied to capture differences in thermostat authority and social dynamics: O1 lives alone, O2 is a couple, O3 has children at home, and O4 carries no household restriction [30, 33]. O4 has the smallest sample ($n = 677$). 720 Additional strata (multi-adult non-family, student dormitories) would draw from the remaining 10,073 “other” respondents but would fragment already thin subsamples without a clearly defined demand response participation profile.

Table C.1: ATUS 2022–2023 occupancy metrics and sample sizes by stratum. % working 09–17 is the mean weighted probability of being engaged in work activity during the 09:00–17:00 window on weekdays (Appendix A). Location columns split each stratum’s work time (ATUS activity codes 05, minutes weighted by TUFINLWGT, episodes with a valid location code) across the respondent’s home, a workplace, and elsewhere; the three sum to 100%. For O2 and O4 the split rests on the small amount of work time those strata report. The at-home share is the work-from-home probability of Section 2.

ID	Stratum	% working 09–17 (wkdy)	% at workplace	% at home	% else- where	<i>n</i>
O1	Employed single adult	60.9	70.2	24.3	5.6	1,393
O2	Retired couple	0.4	8.1	50.1	41.8	2,351
O3	Employed parent	62.6	68.9	25.7	5.4	2,190
O4	Not-employed adult	2.4	26.6	68.7	4.6	677

Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research

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Abstract

Occupants are a primary source of uncertainty in building energy consumption and management, yet existing occupant behavior models cannot capture adaptive and reasoning responses considering the occupant’s personal history, current context, and the type of energy signal being delivered. This study presents BUILD OCC, an open-source Python platform that grounds large language model agents in the American Time Use Survey (ATUS), a nationally representative diary dataset ~~covering 16,684 respondents~~, using the 6,611 diary respondents who fall within four demographic strata. Through BuildOcc, each simulated occupant agent can be instantiated with a demographic persona drawn from ATUS population statistics, a memory stream that accumulates and reflects on timestep-level observations, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface—Python library, REST ~~API~~application programming interface (API), and Model Context Protocol server—so that any building energy tool (EnergyPlus, Home Assistant) can integrate behavioral intelligence without bespoke coupling code. A plugin registry lets the community add new occupant strata, custom schedulers, and alternative memory backends as separate installable packages. Two validation tiers show that ATUS-grounded sampling reproduces ~~empirically-calibrated activity distributions~~its ATUS reference distributions to within sampling noise and that demographic priors propagate into persona-consistent agent reasoning across timesteps, establishing internal consistency across strata. BUILD OCC provides the building energy community with a reusable, openly available implementation of the occupant behavioral layer. BUILD OCC is openly released at <https://doi.org/10.5281/zenodo.22255938> under the Apache License 2.0 and installable via `pip install buildocc`.

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Keywords: Artificial Intelligence Agent, Occupant Behavior Modeling, Building Energy Use and Management, American Time Use Survey

Required Metadata

Current code version

Table 1: Code metadata (mandatory)

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.1
C2	Permanent link to code/repository	https://github.com/humanbuildingsynergy/Build0cc
C3	Permanent link to reproducible capsule	https://doi.org/10.5281/zenodo.22255938
C4	Legal code license	Apache License 2.0
C5	Code versioning system	Git
C6	Software code languages, tools, and services used	Python 3.11+
C7	Compilation requirements, operating environments & dependencies	<code>pip install build0cc</code> ; requires a large language model (LLM) API key (Anthropic, OpenAI, Google), or local Ollama
C8	Developer documentation	https://github.com/humanbuildingsynergy/Build0cc/blob/main/README.md
C9	Support email for questions	wooyoungjung@arizona.edu

1. Motivation and Significance

Humans are complex systems that respond to environmental, social, and economic signals in ways that are difficult to predict, introducing substantial uncertainty in building energy use and management [1, 2][1, 2, 3]. This long-standing challenge has motivated the development of occupant behavior models that can capture the ~~stochastic-unpredictable~~ and adaptive nature of human decision-making [4]. However, most existing models rely on fixed schedules (e.g., ASHRAE 90.1 reference schedules [5])~~or stochastic processes that~~

do not comprehensively capture and reason about the occupant’s personal history, current context, or the type of energy signal being delivered [4, 2]. Consequently, these models cannot replicate the nuanced and adaptive responses that real occupants exhibit in response to various conditions. [5, 6]

15), which assign the same profile to every day and every occupant. Stochastic models improved on this by sampling from sensor data [7] or national time-use diaries: early work used first-order Markov chains to generate occupant behaviors with realistic day-to-day variation [8, 9]; later models added explicit behavior episodes with durations [10] and higher-order transitions that distinguish

20 occupant types [11]. Each step reproduces occupant behavior statistics more faithfully, but all remain statistical descriptions.

The emergence of large Large language models (LLMs) ~~and their architecture~~, with architectures for reasoning and memory ~~has opened new opportunities for artificial intelligence (AI)~~, allow agents to simulate human-like decision-

25 making processes. ~~These agents are capable of~~ through contextual reasoning, memory retention, and adaptive behavior, ~~making them suitable for modeling complex systems, including behavior—capabilities suited to modeling~~ occupants in buildings. Recent studies demonstrated the potential of LLM-integrated agents to simulate occupant behavior in building energy contexts,

30 showing that they can produce more realistic and adaptive responses to various energy signals compared to traditional models [4, 12, 13, 14, 15, 16]. Specifically, Nicholas et al. [17] proposed agents that adopt the generative agent memory architecture of Park et al. [18]—combining a persona-based core memory, a timestep-level memory stream, and periodic reflection—

35 and demonstrate that this representation produces qualitatively different energy demand patterns than fixed-schedule or rule-based occupant models. Other ~~recent approaches include agents conditioned on IEQ sensor readings for comfort and health feedback~~ agents extend this architecture in different directions: conditioning behavior on indoor environmental quality [19], ~~personality-driven~~

40 ~~agents for negotiating shared thermostat setpoints~~ negotiating a shared setpoint between occupants [20], ~~and a Thought-Action-Reflection-Journal framework for modeling energy behavior under~~ responding to dynamic pricing [21], and adjusting setpoints under demand response [22]. However, these agents are instantiated from hand-crafted profiles or small convenience samples [23].

45 ~~In addition, few provide quantitative validation against empirical activity benchmarks, few ground their schedules in national survey data, and few release an open-source platform for integration with building energy tools—leaving the community without a~~ are released as open platforms—leaving no shared, survey-grounded starting point for reproducible occupant behavior research.

50 This study ~~aims to fill this gap by introducing~~ addresses that gap with BUILD-OCC, an open-source platform for LLM-based occupant ~~behavior simulation~~ and

building energy management. BUILD OCC grounds agent personas simulation,
Personas and activity schedules are grounded in the American Time Use
Survey (ATUS) [24], ~~a nationally representative dataset covering tens of~~
55 ~~thousands of diary respondents across income, employment, and household-type~~
~~strata.~~ Each agent maintains and each agent carries a memory stream
that accumulates ~~timestep-level~~ observations and periodically ~~synthesizes~~
~~higher-order behavioral insights through reflection—enabling coherent, adaptive~~
~~decision-making reflects on them, giving coherent behavior~~ across a simu-
60 ~~lation day rather than independent per-timestep responses.~~ Users can drive
agents through building simulation timesteps using any supported LLM provider
and receive structured behavioral decisions—thermostat adjustments, device
control actions, and demand response reactions—that feed directly into
~~A three-layer interface (Python library, REST API, and Model Context~~
65 ~~Protocol (MCP) server) connects the agent to~~ EnergyPlus [25] co-simulation,
Home Assistant ~~automations, or, or a~~ custom building management ~~systems~~
~~through a three-layer interface (Python library, REST API, and MCP server).~~
~~Abstract base classes and system, and~~ a plugin registry ~~allow the community~~
~~to extend the platform with new occupant strata, behavioral data~~ lets other
70 groups add strata, grounding sources, and memory architectures. ~~Consequently,~~
~~BUILD OCC offers a reusable and extensible implementation of the occupant~~
~~behavioral layer, released openly so that it can be evaluated and built upon~~
~~by other research groups.~~

2. Software Description

75 Following an overview of the platform architecture, the platform is described
in six parts: the ATUS behavioral grounding that drives activity scheduling
and persona sampling; the inputs required at simulation initialization; the
per-timestep LLM reasoning pipeline; the demand response signal interface;
the extensibility mechanisms that allow the platform to be adapted to new
80 research contexts; and the computational cost of running it.

2.1. Architecture Overview

BUILD OCC is organized around two levels of architecture: an internal *agent*
component architecture that defines how occupant agents perceive, reason,
and act within a simulation timestep, and a *platform interface* that exposes
85 this functionality to external building energy tools (Figure 1).

2.1.1. Agent Component Architecture

The BUILD OCC occupant agent is composed of six interacting components:

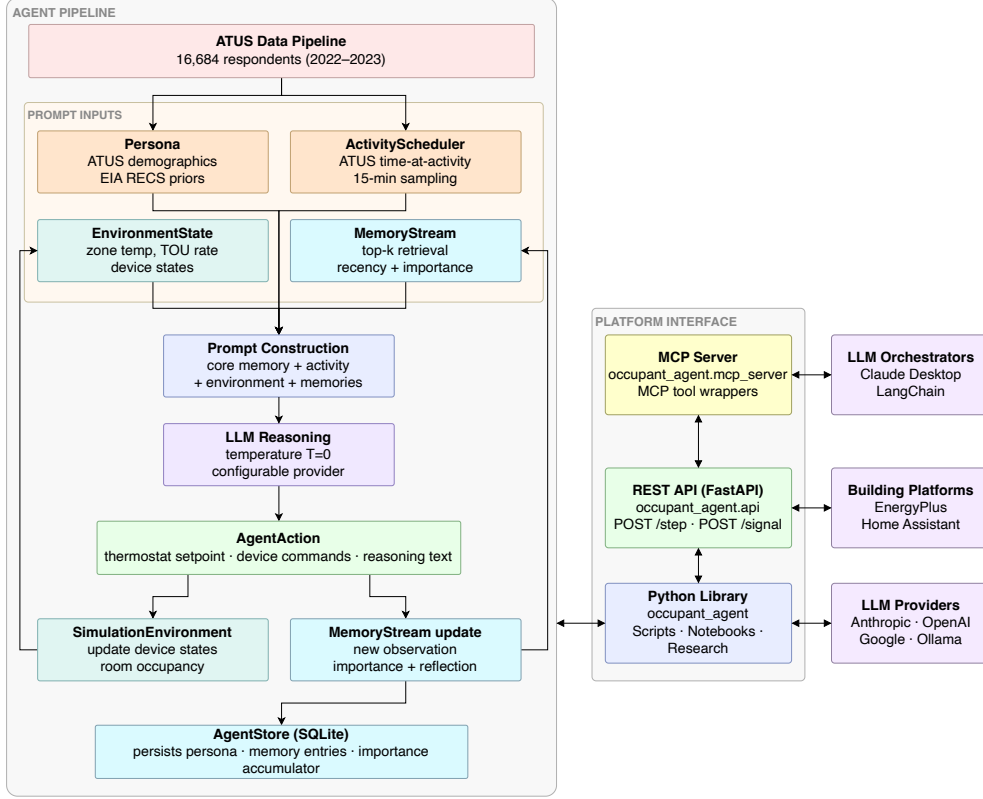


Figure 1: Architecture of BUILD OCC. *Left*: per-timestep agent pipeline showing how ATUS-grounded activity codes, retrieved memories, and environment state are assembled into an LLM prompt and how the resulting **AgentAction** updates both the simulation environment and the memory stream. *Right*: three-layer platform interface through which external building energy tools and LLM orchestration frameworks access the agent.

- Persona** — a structured demographic profile assigned to one of four strata (O1–O4; Table 2), with appliance ownership ~~drawn from Residential Energy Consumption Survey (RECS) [26] priors, an income bracket drawn uniformly within the stratum’s RECS range, and a comfort band set from income bracket, and thermal comfort tolerance sampled from RECS [26] probability distributions.~~
- ActivityScheduler** — samples ~~the occupant’s activity at each 15-minute timestep from stratum- and hour-specific probability distributions derived from ATUS 2022–2023 diary records (16,684 respondents), with weekday/weekend stratification (Appendix A, hour- and day-type-specific ATUS distributions (6,611 respondents; Appendix C).~~

- 100 • **MemoryStream** — an accumulating log of observations, actions, and reflections, ranked by recency and LLM-assigned importance following Park et al. [18] ~~; top- k entries are injected into each timestep prompt and periodically synthesized into higher-order behavioral insights through a reflection call~~ (Appendix B).
- 105 • **LLM Reasoning Engine** — ~~assembles a prompt from the persona description, current ATUS activity, environment state, and retrieved memories, then~~ queries a configurable LLM provider (Anthropic, OpenAI, Google, or local Ollama) at temperature $T=0$ ~~for reproducibility~~, returning a structured action with a natural-language reasoning string.
- 110 • **SimulationEnvironment** — maintains device states, room occupancy, and thermostat setpoint between timesteps: ~~at the start of each timestep it exposes the current state to the LLM Reasoning Engine, and at the end it applies the agent's returned action,~~ exposing the current state before each decision and applying the returned action after it.
- 115 • **AgentStore** — persists the agent's cognitive state (persona, ~~memory entries, and the~~ memories, importance accumulator) to SQLite ~~via SQLAlchemy~~ after each operation. Environment state is ~~owned by the caller and passed fresh with each request, enabling a stateless REST API pattern that supports multi-process deployment~~ not stored: the calling program sends it with every request, so the server keeps nothing
120 between calls.

2.1.2. Platform Interface

BUILD OCC is distributed as the package `builddocc` and imported as `occupant_agent`. It exposes the agent through three integration layers, each targeting a different class of integrator:

- 125 1. **Python library** (`occupant_agent`) — ~~in-process~~ direct access to all agent components ~~with no server overhead~~; the recommended path for batch simulations ~~and notebook experiments~~.
- 130 2. **REST API** (`occupant_agent.api`) — a stateless web server (FastAPI [27] ~~server~~) that keeps no state between requests, exposing agent initialization, per-timestep decisions (POST `/step`), signal delivery (POST `/signal`), and state query (GET `/state`); suitable for EnergyPlus co-simulation callbacks and any platform that can issue ~~HTTP~~ ordinary web (HTTP) requests.
- 135 3. **MCP server** (`occupant_agent.mcp_server`) — wraps the REST API as Model Context Protocol [28] tools, so any MCP-compatible host

(Claude Desktop, LangChain, Home Assistant) can drive a simulation without additional integration code.

2.2. ATUS Behavioral Grounding

Activity schedules are derived from ATUS 2022–2023 microdata (~~16,684 respondents~~).
 140 Time-at-activity probabilities are aggregated across ~~eight~~ nine categories (sleeping, work, food preparation, laundry, television viewing, eating, exercise, travel, and other) ~~at 15-minute resolution~~ by hour of day, stratified by demographic group and day type (weekday / weekend). The method follows the “what is the person doing at this moment?” framing: for each
 145 diary episode $[t_{\text{start}}, t_{\text{stop}}]$, the method queries whether the episode covers the half-hour mark of each hour h , then computes a weighted probability across respondents using ~~ATUS person-level weights. ATUS records a single primary activity per episode, so simultaneous secondary activities—watching television while preparing food, for example—are not represented in the scheduler’s output~~
 150 the ATUS final person weight (TUFINLWGT). Weighting makes each probability an estimate for the U.S. population, and because the weight is defined identically in the 2022 and 2023 files the two survey years can be pooled. The formal computation is given in Appendix A. The scheduler draws each timestep independently from the hourly distribution, so it reproduces the hourly marginals without modeling how long an activity lasts, and episode lengths are consequently shorter than the diaries record (Section 3.2); no duration model is implemented in this release.
 155 Four demographic strata are defined (Table 2), selected to span the variation in building occupancy and demand response participation that is most relevant to residential energy research (Appendix C). Of the 16,684 respondents surveyed, 6,611 fall within these four strata.

Table 2: Demographic strata

ID	Label	ATUS filter
O1	Employed single adult	Full-time employed, lives alone, age 25–44
O2	Retired couple	Age 65+, not employed, spouse or partner present
O3	Employed parent	Full-time employed, children present, age 35–54
O4	Not-employed adult	Not employed (unemployed or not in the labor force), age 25–44

~~An important methodological choice is the *in-progress* framing: rather than sampling from the start-time distribution (which biases toward short activities), the~~

method evaluates which activity is in progress at the half-hour mark of each hour, reflecting the actual time spent in each activity state rather than the frequency of starts. The formal probability computation is described in Appendix A.

Work-from-home (WFH) behavior is handled separately: a per-agent WFH probability encodes the fraction of working days spent at home (~~drawn from ATUS 2022–2023 telework questions~~ the share of each stratum’s work time that ATUS respondents report at home; Table C.1), and a per-day Bernoulli draw determines WFH status for each simulated day. The WFH flag is included in the agent’s per-timestep prompt so that the LLM reflects both the ATUS activity code and the work location, preventing the scheduling model from assigning “work” activities that place the agent outside the home on days it is working remotely.

Each stratum uses stratum-specific appliance ownership priors from ~~EIA the U.S. Energy Information Administration’s~~ RECS [26, 29] and ~~an income-bracket-modulated comfort tolerance~~ a comfort band (in °C) ~~that widens for lower income brackets~~, reflecting the documented tendency of lower-income households to accept wider thermal deviations to reduce ~~HVAC heating, ventilation, and air-conditioning (HVAC)~~ costs [30, 26, 31, 32]. ~~Income brackets are currently sampled uniformly within each stratum’s RECS income range.~~

2.3. User-Provided Inputs

Table 3 lists all inputs required at initialization; the only per-timestep value the calling application must supply is zone temperature, since it depends on the building’s physical response to occupant actions. To lower the barrier to first use, the library ships preset helpers for a representative household device inventory and room layout. A command-line interface ~~also ships preset helpers and a~~ YAML-driven configuration path offers the same presets and a plain-text (YAML) configuration file for users who prefer not to write Python. ~~Users without a live co-simulation link can supply a constant zone temperature (suitable where thermal dynamics are not the focus)~~ Without a live link, zone temperature can be a constant or a pre-computed ~~CSV schedule indexed by datetime; a sample file covering one week of EnergyPlus output is included in the repository~~ comma-separated values (CSV) file; a one-week EnergyPlus sample is included.

2.4. LLM Reasoning Pipeline

~~With all inputs initialized, at~~ At each 15-minute ~~simulation~~ timestep the agent ~~executes the following five-step pipeline:~~
Activity resolution: ~~the~~ resolves its current ATUS code ~~is translated~~ to a natural-language ~~description~~ activity and an occupancy flag ~~using a~~

Table 3: User-provided inputs to BUILD OCC. All are specified once at initialization, except zone temperature, which the calling platform supplies at each timestep.

Input	Type	Description
Stratum	String	Demographic profile: O1–O4
Seed	Integer	Controls persona attribute sampling, activity scheduling, and work-from-home day draws; the same seed produces an identical agent across runs.
LLM provider	String	Anthropic, OpenAI, Google, or Ollama
Device inventory	List	Appliance IDs, initial on/off states, rated power (W)
Room list	List	Rooms the occupant can occupy
Thermostat setpoint	Float	Initial comfort setpoint (°C)
Zone temperature	Source	Fixed value, CSV file, or live building platform
Outdoor temperature	Source	Fixed value, CSV file, EnergyPlus weather (EPW) file, or callable
Time-of-use (TOU) rate schedule	Source	Fixed value, CSV file, or tariff function

lookup table of ATUS activity codes. **Memory retrieval:** (a sleeping or laundry activity moves the agent to the matching room by rule, without a model call), retrieves up to five entries are selected by a retrieval score that combines recency (exponential decay, half-life 24 h) and a 1–10 importance score the LLM assigns to each entry at the time of recording—ensuring that consequential events remain retrievable even as they age (see memories ranked by recency and importance (Appendix B), following Park et al. [18]). **Prompt construction:** a system prompt encodes the persona’s stable identity—demographics, appliance ownership, and comfort tolerance. A user prompt encodes the current environment state, ATUS activity, and the retrieved memories. **LLM call:** the prompts sent to a configurable provider (Anthropic Claude, OpenAI GPT, Google Gemini, or Ollama) at temperature $T=0$ for reproducibility and assembles these with the persona and the environment state into a system and user prompt. The model selects from returns one of four action types (Table 4), returning a JSON object with target, as a structured-text (JSON) object carrying a target, a value, and a one-sentence memory note. **Memory update:** the observation and its LLM-assigned importance score are, which is appended to the stream. If the cumulative accumulator with an LLM-assigned importance score. When the cumulative importance exceeds a threshold (default 100), a reflection pass is

225 ~~triggered in which a higher-capability model~~ distills recent observations into
three higher-order insights~~stored as reflection entries.~~ Figure 1 shows the
same sequence.

Table 4: The four action types returned by `step()` at each timestep, with their decision trigger and a representative example. Demand response is handled separately via `receive_signal()` (Section 2.5).

Action type	Trigger	Example
<code>do_nothing</code>	No environmental or activity condition warrants intervention	Agent is sleeping; no device is running
<code>adjust_thermostat</code>	Zone temperature leaves the comfort band, or TOU rate context	Raises setpoint during peak window to reduce HVAC cost
<code>toggle_device</code>	Activity–appliance alignment (on) or inactivity (off)	Turns off lights when leaving for work
<code>move_room</code>	Activity context suggests transitioning to a different room	Moves to bedroom at 23:00 for sleeping

230 ~~Setting the sampling temperature to~~ The engine samples at $T=0$ makes
~~the model return by default, so the model returns~~ its single most proba-
ble output ~~rather than sampling among alternatives, so and~~ a fixed provider,
model version, and prompt reproduce the same action on every run. ~~This~~
~~determinism does not extend across providers;~~ That determinism is relative
to one set of model weights: it does not hold across providers or across suc-
cessive versions of a hosted model, ~~because it is relative to one specific set~~
~~of model weights.~~ ~~Reproducible studies therefore record the provider and~~
~~model version alongside the seed, and a locally hosted open-weight model~~
235 ~~can be pinned where results must remain.~~ Studies that must stay repro-
ducible over time can run a fixed, locally hosted open model, whose weights
do not change.

240 ~~The pipeline above governs autonomous per-timestep behavior; demand response~~
~~scenarios additionally require the ability to deliver external signals to the~~
~~agent, which is handled by the interface described in the following subsection.~~

2.5. Demand Response Signal Interface

Alongside the per-timestep pipeline, BUILD OCC provides a dedicated signal delivery interface that can be invoked at any point in a simulation. Three

245 signal types are supported (Table 5), following established behavioral inter-
vention typologies [30, 33, 34]. Signals are delivered by passing a signal type,
a free-text message, and the current environment state to the agent, which
returns a response code (*accepted*, *rejected*, or *deferred*) alongside a natural-
250 language reasoning string drawn from the agent’s memory context. Keeping
the three types mutually exclusive enables factorial experimental designs in
which signal type and demographic stratum are the primary independent
variables.

Table 5: Demand response signal types supported by BUILD OCC, classified by behavioral lever and illustrated with representative example messages. All three types share the same delivery mechanism and response codes (*accepted*, *rejected*, *deferred*).

Type	Behavioral lever	Example message
A — Direct command	Regulatory compliance	“Turn off your HVAC for 30 min to support grid stability during today’s peak demand event.”
B — Educational	Economic information	“Your HVAC costs 3 times more per kWh at the current peak rate — raising the setpoint 1°C saves an estimated \$0.35 today.”
C — Social norm	Peer comparison	“75% of similar households in your area have reduced HVAC use during today’s peak event.”

~~The extensibility mechanisms that allow all of these components to be adapted for new research contexts are described in the following subsection.~~

255 2.6. Platform Extensibility

BUILD OCC is designed for adaptation across research contexts without modifying the core library. Extensibility is organized along three independent axes:

- 260 • **Occupant population** (`BasePersona`): new demographic profiles — commercial occupants, hotel guests, non-US populations, or any custom stratum — are registered by implementing `BasePersona`.
- **Activity grounding** (`BaseScheduler`): the built-in scheduler uses ATUS microdata; any other grounding source (a Markov chain, a physics-based occupancy model, or alternative activity data) can be substituted
265 by implementing `BaseScheduler`.

- **Memory architecture** (`BaseMemoryStream`): the built-in recency-plus-importance retrieval can be replaced with ~~embedding-based retrieval~~ retrieval by semantic similarity (embeddings) or any other scheme by subclassing `BaseMemoryStream`.

270 Extensions ~~are registered via Python’s package entry-point mechanism and discovered automatically at import time, allowing third-party packages to extend the platform without forking the repository~~ declare themselves in their own package configuration (Listing 1) and are found automatically when the library loads, so a third party can add a stratum or scheduler without modifying BUILD OCC’s own code:

Listing 1: `pyproject.toml` entry-point declarations for a third-party stratum and scheduler.

```
1 [project.entry-points."occupant_agent.strata"]
2 P5 = "my_package.personas:LowIncomeElderlyAlone"
3
280 4 [project.entry-points."occupant_agent.schedulers"]
5 homer = "my_package.schedulers:HomerScheduler"
```

Transfer to another country is a data-substitution task rather than a re-design: the scheduler consumes a table of activity probabilities by stratum, hour, and day type, so any national time-use survey following harmonized diary ~~conventions—the conventions~~ [35]—the Multinational Time Use Study [36] or the European HETUS series Harmonised European Time Use Surveys (HETUS) [37], for example—can be substituted by regenerating that table behind `BaseScheduler`. What does not transfer automatically is the persona layer, whose appliance-ownership and income priors come from a US survey (RECS) and would need a national equivalent.

290 ~~Researchers can inject additional context into the LLM reasoning pipeline—such as predicted mean vote (PMV) comfort~~ Additional context—comfort indices [38], language-based comfort and heat-strain signals [39, 40], or ~~indoor air quality readings—either air quality readings—can be injected~~ by extending the agent class or by passing ~~a free-text string~~ free text at each timestep; ~~both paths are available through the REST API and MCP server, through either interface.~~ The environment ~~state~~ and signal schemas are fixed ~~to ensure for~~ reproducibility, and a ~~test utilities module provides mock implementations and conformance tests for extension authors testing module lets extension authors check, without an API key, that a new stratum or scheduler meets the platform’s contract.~~

2.7. Computational Cost and Scalability

Because the agent queries a language model at ~~nearly every timestep~~ most timesteps, cost and runtime scale linearly with simulated agent-days. A 15-

minute timestep gives ~~up to at most~~ 96 calls per simulated day per agent, plus occasional reflection calls, with responses capped at 512 tokens; ~~the monetary cost therefore follows~~. Measured over two simulated agent-days for each of the four strata (September 2026 list prices), one agent-day required 85.5–90.0 per-timestep calls and 1.5 reflection calls, consumed 111,000–122,000 prompt and 13,700–15,300 completion tokens, and cost \$0.19–\$0.21. Fewer than 96 calls are issued because room transitions that follow directly from the provider’s per-token price at the time of use, and wall-clock time sampled activity, such as moving to the bedroom when sleep begins, are resolved without a model call; they are still counted as moves in Table 7. Wall-clock time was 2.1–4.3s per timestep and is dominated by provider latency rather than, not local computation. Two mechanisms reduce this cost. Any local open-weight model served through Ollama removes per-call cost entirely, at the price of reduced reasoning quality, and scheduler-only studies—including the Tier 1 validation reported in Section 3.2—require no model calls at all. Cost, rather than platform capability, is what bounded the simulation lengths reported in Section 3.

These figures are per agent. ~~Because agents~~ Agents are independent—each carries its own persona, memory, and schedule, and the REST API is ~~stateless—a~~ stateless—so a multi-occupant household is simulated by running runs one agent per occupant, so call volume scales linearly with household size (a four-occupant household is roughly 384 calls per simulated day) while the agents themselves can be run in parallel across processes. Interaction between occupants, such as negotiation over a shared thermostat, is not modeled in this version and is identified as future work in Section 5. with call volume scaling linearly and the agents running in parallel.

3. Illustrative Examples

This section demonstrates the platform in three parts: a walkthrough of a single agent through a demand response event, followed by two quantitative tiers. Tier 1 validates the *ActivityScheduler*, testing whether ATUS-grounded sampling produces category distributions matching empirical ATUS reference data relative to a deterministic rule-based baseline; Tier 2 validates the *LLM agent decisions*, testing whether each stratum’s demographic priors propagate into distinguishable behavioral patterns under identical environment conditions. Both are necessary because a well-grounded scheduler and a well-calibrated reasoning agent are independent failure modes — one can be correct while the other is not. All LLM results in this section use Anthropic models at $T=0$ with a 512-token response cap: `claude-haiku-4-5-20251001`

345 for per-timestep decisions and [claude-sonnet-4-6](#) for reflection synthesis;
Tier 1 involves no model calls.

3.1. Demand ~~response walkthrough~~ Response Walkthrough

To illustrate agent behavior during a residential demand response event, this subsection traces an O1 agent (28-year-old employed male, income bracket 9 on the 16-level RECS scale, comfort band $\pm 1.1^\circ\text{C}$ around the setpoint)
350 through the peak electricity period of a summer weekday evening (18:00–19:45, TOU rate \$0.22/kWh). Zone temperatures come from pre-computed EnergyPlus output ~~for the DOE PNNL residential prototype building [41] (single-family, Climate Zone 2B, variable-speed heat pump)~~ included in the repository [\[41\]](#); outdoor temperature follows a synthetic summer-day profile.
355 The simulation loop is:

Listing 2: Running the BuildOcc simulation loop with the sample zone temperature CSV.

```
1 # Full imports available in examples/simulation_loop.py (
    repository)
2 from datetime import datetime, timedelta
360 3
4 agent      = OccupantAgent.from_stratum("01", seed=42)
5 scheduler = ActivityScheduler(stratum="01", seed=42)
6 zone_temp = zone_temp_from_csv(...) # EnergyPlus zone
    temperatures
365 7 price_message = ("Your HVAC is running during peak hours (
    4-9pm) at $0.22/kWh -
    8
        "three times the off-peak rate. Raising
        the setpoint by 1°C saves ~$0.35
        today.")
370 9
10 sim = SimulationEnvironment(
11     initial_devices=persona_devices(agent.persona),
12     initial_rooms=[RoomState(room_id="living_room",
        occupied=True)],
375 3     thermostat_setpoint=22.0,
14     outdoor_temp_fn=summer_day_temp,
15     tou_rate_fn=peak_tou_rate,
16 )
17
380 8 base = datetime(2025, 8, 11, 18, 0)
19 for i in range(8):
20     ts      = base + timedelta(minutes=15 * i)
21     env      = sim.observe(ts, zone_temp(ts))
22     if i == 2:
```

```

385:3         agent.receive_signal("B", price_message, env)
24         action = agent.step(env, atus_code=scheduler.sample(
            ts))
25         sim.apply(action, ts)
26         print(ts.strftime("%H:%M"), action.action_type,
390         action.reasoning)

```

At 18:00 (Step 1), the ~~CSV-sourced zone temperature zone~~ is 23.9°C and the thermostat is set to ~~against a~~ 22°C. ~~With outdoor temperatures exceeding 38~~ setpoint, with 35°C, the HVAC is running continuously to hold the setpoint. ~~The agent is engaged in computer leisure and proactively raises the thermostat setpoint to 24~~ outdoors and the peak rate in force. The sampled activity is work-related travel, so the agent is on its way home and raises the setpoint to 23.5°C ~~to reduce peak-rate HVAC run time, reasoning:~~

~~“The indoor temp is 23.9°C and the thermostat is set to 22°C —the HVAC is running hard to maintain the setpoint with 39°C outside—I’m away from home~~ during peak rate hours (\$0.22/kWh). ~~I’m comfortable at 23.9°C, well within my 1.1°C tolerance,—so raising the setpoint to 24~~ 23.5°C will meaningfully cut cooling costs during reduce HVAC runtime and save money while I’m not there to notice the slight warmth. The current 1.9°C deviation is already outside my comfort band, so this ~~expensive period~~ adjustment is justified both for cost and to prevent further temperature drift.”

~~This response demonstrates that the persona’s income bracket and comfort tolerance surface in actionable decision-making without explicit rule encoding.~~

At 18:15 the agent arrives and switches on the television. At 18:30 (Step 3) ~~;~~ a Type B educational signal is delivered: “Your HVAC is running during peak hours (4–9pm) at \$0.22/kWh—three times the off-peak rate. Raising the setpoint by 1°C saves ~\$0.35 today.” The agent ~~accepts the signal but references declines, on the strength of~~ its own prior action ~~stored in memory:~~

~~“I already raised the thermostat to 24°C earlier today and it’s working—I’m comfortable right now at 23.9°C and comfortable within my tolerance~~ just settled in to watch TV. The math is clear Raising the setpoint by 1°C would push me toward discomfort (my limit is ±1.1°C from setpoint), and \$0.35 savings ~~—isn’t worth feeling warm and uncomfortable for the effort is zero since I’ve already taken this action~~ next few hours.”

~~This~~ For the next hour the agent holds the setpoint, recalling at each step that it adjusted the thermostat a short time ago; at 19:30, when the sampled activity turns to sleep, it switches the television off. This cross-timestep coherence—the agent recalling and acting on its own prior decision—is coherence—recalling an earlier decision and judging a new request against it—is what a fixed schedule cannot represent.

~~Over the remainder of the evening the agent maintains the raised setpoint without further intervention; importance ratings for routine observations stay low (2–3 out of 10)~~ The zone temperature comes from the pre-computed file and does not respond to the setpoint change, so the walkthrough is open-loop; closing that loop is the role of the co-simulation interface (Section 2.1.2).

3.2. Tier 1: ~~Activity Schedule Fidelity~~ *Sampler Correctness*

The ATUS-grounded ~~activity scheduler is validated against~~ scheduler is compared with a deterministic rule-based baseline using KL divergence between simulated activity distributions and ATUS empirical distributions by Kullback–Leibler (KL) divergence, which measures how far the simulated distribution departs from the reference. For each stratum, 180 days of 96 timesteps each (17,280 observations per stratum) are simulated; the empirical category distribution at each hour is computed, and compared to and the hourly category distribution is scored against the ATUS reference using:

$$D_{\text{KL}}(P\|Q) = \sum_c P(c) \log \frac{P(c)}{Q(c) + \varepsilon} \quad (1)$$

where P is the ATUS reference distribution, Q is the simulated distribution, and $\varepsilon = 10^{-9}$ is a smoothing constant. Table 6 and Figure 2 report the mean per-hour KL divergence averaged across all 24 hours and both day types over the weekday distribution, which places 520 simulated timesteps in each hourly cell. The scheduler samples from the same table it is scored against, so the benchmark for a correct sampler is not zero: at finite n the realized distribution departs from P by chance alone. Table 6 therefore also reports a null obtained by drawing 520 samples per hour directly from P and scoring them with Equation 1, over 1,000 repetitions.

Every ATUS-grounded condition falls inside or below its sampling null, so the scheduler reproduces its reference tables to within sampling noise. The fixed baseline, which assigns deterministic activities by hour-of-day rules (e.g., sleeping 23:00–06:59, work 09:00–16:59) without demographic conditioning, which produces large divergence against the ATUS reference—especially diverges two orders of magnitude more, most of all for the retired (O2) and unemployed not-employed (O4) strata, whose schedules differ most depart furthest from

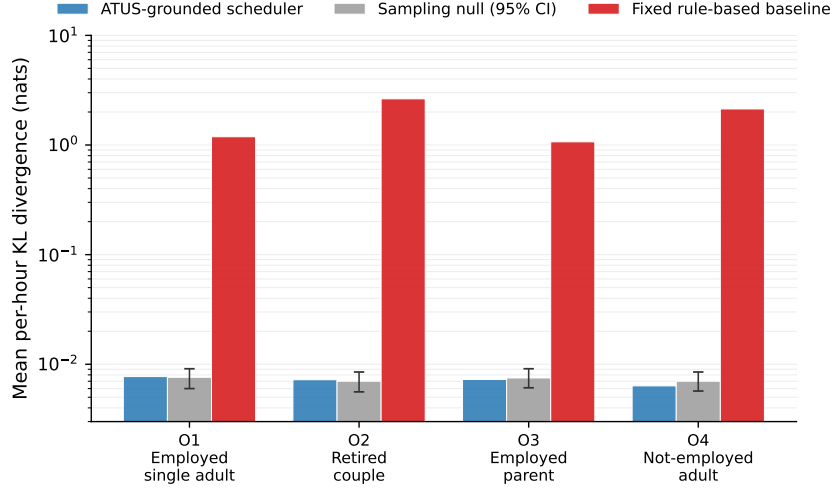


Figure 2: Mean per-hour KL divergence (nats, log scale) between simulated and ATUS empirical activity distributions over 180 simulated days, weekdays. The baseline’s magnitude is governed by ATUS-grounded conditions sit at the smoothing constant, since it places zero probability on categories the reference records as occurring sampling-noise floor for this sample size (Table 6).

Table 6: Tier 1 sampler correctness. Mean per-hour KL divergence (nats, natural-log units) against the ATUS reference, over 180 simulated days of 96 timesteps (520 weekday observations per hourly cell). The null is the divergence expected from a correct sampler at the same n , with its 95% confidence interval (CI), drawn directly from the reference distribution over 1,000 repetitions. Lower is better.

Stratum	ATUS scheduler	Sampling null (95% CI)	Fixed baseline
O1 (Employed single)	0.0077	0.0076 (0.0060–0.0091)	1.19
O2 (Retired couple)	0.0072	0.0070 (0.0056–0.0085)	2.64
O3 (Employed parent)	0.0073	0.0075 (0.0061–0.0091)	1.07
O4 (Not-employed)	0.0064	0.0070 (0.0057–0.0085)	2.13

the employed-adult assumption. Every ATUS-grounded condition stays below 0.03 nats, while the fixed baseline ranges from 9.92 to 11.48. The gap is one of kind rather than degree: the baseline places zero probability on categories the reference records as occurring, so its divergence is governed by the smoothing constant ϵ and registers the presence of impossible predictions rather than the size of a behavioral error.

Figure 3 illustrates what this divergence looks like shows the contrast category by category for stratum O1 on weekdays: the ATUS-grounded sim-

ulation (panel a) closely tracks the reference distributions, while the ~~fixed rule-based~~ baseline (panel b) ~~diverges markedly—most notably collapsing~~ collapses daytime activity into sharp step functions and ~~failing to reproduce~~ misses the gradual evening wind-down.

Per-hour marginals do not constrain episode length, however. Because each timestep is drawn independently, episode lengths are geometrically distributed by construction, each step ending the episode with a fixed probability: simulated sleep episodes average 52–71 min across strata against 314–338 min in the diaries, and agents make 49–54 activity transitions per simulated day.

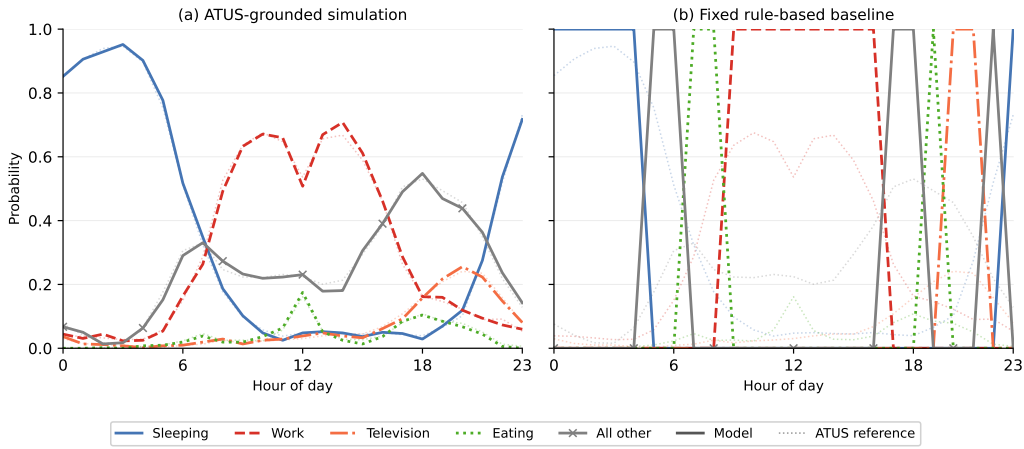


Figure 3: Hourly activity probability distributions for stratum O1 (employed single adult, weekday) ~~across five activity categories~~. ~~Four load-relevant categories are drawn individually (asleeping, work, television, and eating) ATUS-grounded simulation (and the remaining five are pooled as “all other”).~~ Curves are population probabilities over the weekdays of 180 simulated days (520 timesteps per hour, as in Table 6); an individual diary day is a step function, and the smoothness here is a property of that average. (a) ~~ATUS-grounded simulation~~: bold lines ~~closely~~ track the ~~dotted~~ reference. (b) Fixed rule-based baseline: bold lines ~~diverge sharply, with depart from the baseline collapsing smooth activity transitions into unrealistic step functions reference and misassigning misassign~~ the O1 late-evening ~~activity~~ peak.

3.3. Tier 2: Persona-Driven Behavioral Differentiation

Tier 2 tests whether ~~the demographic priors encoded in~~ each stratum’s ~~persona produce distinguishable behavioral patterns under identical environment conditions, so that any observed differences are attributable to persona variation rather~~ than environment variation. ~~Each of demographic priors produce distinguishable behavior.~~ O1–O4 ~~was were~~ each run for three ~~independent~~ seeds over one weekday (96 timesteps ~~at 15-minute resolution~~) under identical ~~environment~~

conditions—the same ~~EnergyPlus~~ zone temperature, device configuration (~~HVAC~~, ~~three lighting zones~~, ~~TV~~, ~~washer~~, ~~dishwasher~~), and time-of-use (TOU) tariff (\$0.22/kWh peak, \$0.08/kWh off-peak) so that any difference is attributable to the persona.

Table 7 reports ~~the resulting~~ action-type distributions and mean thermostat setpoints during the TOU peak window (4–9 pm). and Figure 4 visualizes the active-action counts. The ordering follows daytime home occupancy: ~~O2 (retired couple),~~ at home all day, is the most active ~~stratum—lowest inactivity (67.4%), highest device-toggling (8.0%) and room-movement (21.5%) rates, and the highest mean peak setpoint (24.3°C)—consistent with full-day home occupancy.~~ stratum, with the fewest inactive timesteps and the most device toggling and room movement, followed by O4 (unemployed adult; ATUS
 ~~$n=107$, Appendix C) matches;~~ the two employed strata are the least active. Of the moves, 17–25 per stratum follow by rule from a sampled sleeping or laundry activity; the remaining 26–44 are model decisions. Every stratum ends the peak window with its setpoint 1.2–1.6°C above the 22°C it started at, ~~O2’s toggling rate but records the lowest thermostat-adjustment rate (1.4%) and peak setpoint (23.1°C).~~ Because O4’s probability tables derive from only 107 ATUS respondents—the smallest of the four strata and the noisiest at 15-minute resolution—this pattern should be read as provisional; it may reflect sampling noise in the grounding data as much as a demographic tendency. O3 (employed parent) shows the lowest device-toggling rate (3.5%), its peak-hour schedule dominated by ATUS child-care codes, while O1 sustains a moderate profile throughout. Figure 4 visualizes these distributions. highest. Thermostat adjustments are rare in every stratum (1.7–2.4% of timesteps), so the strata differ in what they do with devices and rooms, not in how often they touch the thermostat.

Table 7: Action-type distribution (%) per stratum: mean over three seeds, one weekday each (96 steps), giving 288 timesteps per stratum. At this sample size one percentage point is 2.9 timesteps, so differences of a few tenths should not be read as meaningful. Peak SP is the mean thermostat setpoint during TOU peak hours (4–9 pm, \$0.22/kWh).

Stratum	Inactive	Thermostat	Toggle	Move	Peak SP (°C)
O1 (Employed single)	75.3	2.4	6.2	16.0	23.2
O2 (Retired couple)	60.1	2.1	13.9	24.0	23.6
O3 (Employed parent)	77.1	2.1	4.9	16.0	23.4
O4 (Not-employed)	67.7	1.7	11.1	19.4	23.2

To assess demand response participation, three standardized signals were delivered to each stratum (five seeds each) after a ~~2.5-hour context period at peak-rate conditions (10 simulated timesteps beginning at 10-timestep~~

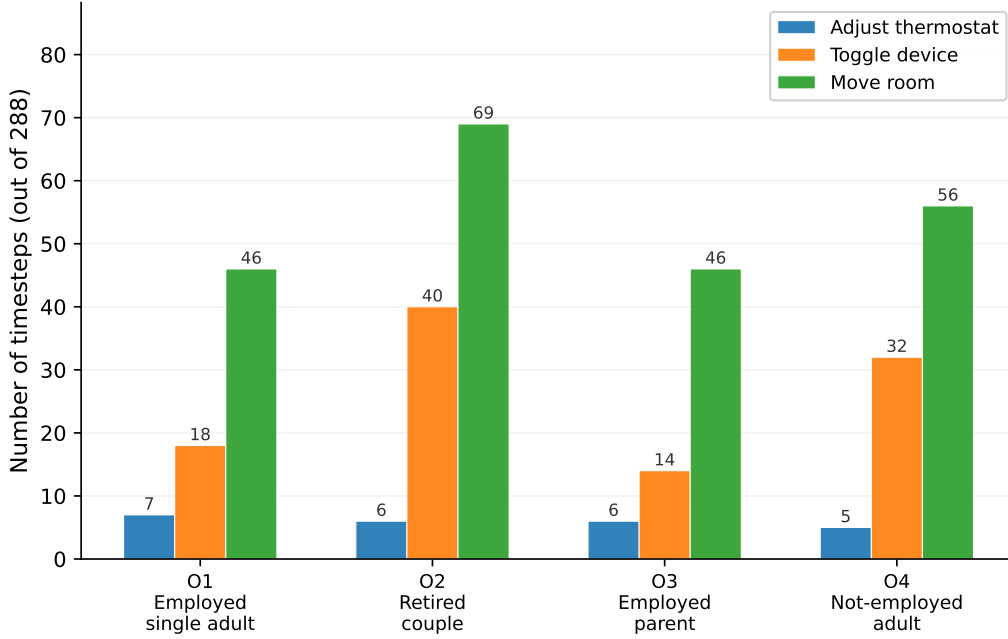


Figure 4: Active action counts per stratum across 288 timesteps (3 seeds $\times$ 96 steps, identical environment conditions). Do-nothing accounts for ~~67–76~~^{60–77}% of all timesteps across strata and is excluded to isolate cross-stratum differences in active decision-making. Bar labels show absolute counts.

~~warm-up from 16:00, giving each agent an experiential baseline in memory before~~
~~signal delivery) at peak rates that seeds each agent’s memory:~~ Type A (direct
515 command: “Turn off your HVAC for 30 minutes”), Type B (educational price
signal: ~~cost comparison at three times the a cost comparison stating the peak~~
~~rate as three times off-peak rate~~), and Type C (social norm: “75% of similar
households have reduced HVAC use”). Table 8 reports the number of accept-
ing agents per stratum and signal type. Direct commands (Type A) ~~drew~~
520 ~~the most acceptance for O2 (retired couple, 2 of 5 seeds), with O1 (employed~~
~~single) and O4 (unemployed adult) accepting in 1 of 5; educational were~~
~~declined in all but one case: the agents reasoned that a 30-minute HVAC~~
~~shutdown on a 35°C afternoon would carry the zone past their comfort band.~~
Educational price signals (Type B) were accepted ~~by O2 and O4 in 2 of 5~~
525 ~~seeds each, a pattern consistent with cost-aware strata responding to both~~
~~authority and economic framing. O3 (employed parent) rejected all signal~~
~~types across all five seeds: signal delivery at 18:30 coincides with child-care~~
~~activity codes in the ATUS schedule, making that outcome an expected~~
~~consequence of schedule grounding rather than a general insensitivity to~~

530 ~~demand response~~ in every stratum, from one agent in five (O3) to three in
 five (O4), and every acceptance cited the concrete dollar figure together with
 headroom inside the comfort band. Social norm signals (Type C) drew no
 acceptance in any stratum, ~~consistent with field~~. Field evidence on the
 limited efficacy of peer-comparison messaging [30, 33]. ~~With five runs per~~
 535 ~~cell these counts indicate direction only; no significance testing is performed~~
~~on them~~ nearest interventions is split: appeals to conserve during repeated
 peak events lose their effect by the third event [42], whereas periodic reports
 comparing a household’s use with its neighbors’ sustain savings of about
 2% [43]. Type C is a single peak-event message, so only the first finding
 540 bears on it.

Table 8: Signal compliance per stratum and signal type, reported as the number of accepting agents out of five random seeds. Percentages are not given, since five runs per cell cannot support them. Type A: direct command; Type B: educational price signal; Type C: social norm nudge. No agent deferred.

Stratum	Type A	Type B	Type C
O1 (Employed single)	0/5	2/5	0/5
O2 (Retired couple)	0/5	2/5	0/5
O3 (Employed parent)	0/5	1/5	0/5
O4 (Not-employed)	1/5	3/5	0/5

These results show that the ATUS demographic priors propagate into ~~qualitatively~~
 distinct LLM-driven behavioral patterns: ~~employment status shapes thermostat~~
~~adjustment frequency, household composition modulates signal compliance,~~
~~and full-day home occupancy drives device toggling behavior~~ daytime home
 545 occupancy drives device toggling and room movement, price information is
the only signal accepted in every stratum, and neither a direct command nor
a social comparison moves an agent that is already inside its comfort band.
 Three constraints bound the interpretation. First, ~~and most fundamentally,~~
 Tier 2 establishes *internal consistency*: agents instantiated from different de-
 550 *mographic priors behave differently, and in directions consistent with those*
priors. It does not establish behavioral realism, which would require compar-
ison against measured occupant data and is left to future work [44]. Second,
 each stratum is run for ~~only~~ one weekday per seed (three seeds for action
 distributions, five for signals), which ~~is sufficient to show directional patterns~~
 555 ~~but not to establish statistical~~ shows direction but cannot establish signif-
 icance; multi-day designs ~~are feasible but~~ were constrained by API cost.
 Third, ~~Tier 2 uses~~ a single environment configuration (~~fixed weather, tariff~~
~~structure, and household composition~~), is used, which limits generalizabil-

ity across ~~different~~ climate zones and occupancy contexts. ~~The behavioral differences across strata also reflect a structural property of the platform: both the ATUS-grounded activity schedule and the LLM persona reasoning are conditioned on the same demographic stratum, a coupling that ensures internal consistency between the scheduling and decision-making components.~~

4. Impact

BUILD OCC lowers the barrier to four lines of research that are difficult to pursue with fixed-schedule occupant models:

Survey-grounded behavioral agents. BUILD OCC packages population-calibrated occupant agents as a reusable component rather than a per-project infrastructure investment. Research groups can introduce demographically differentiated occupant behavior into any building energy study without constructing a grounding pipeline from scratch. The ATUS-derived activity distributions and RECS-conditioned personas provide a documented, survey-grounded baseline that any researcher can adopt, extend, or cite as a reference condition.

Demographic behavioral experiments. BUILD OCC’s combination of ATUS-grounded demographic strata, persona income brackets, and configurable intervention signals enables controlled factorial studies of occupant behavior. Unlike fixed-schedule models, each agent reasons from its demographic persona and accumulated memory at every timestep, producing adaptive responses that reflect personal history and current context rather than predetermined patterns. Because both the activity schedule and the agent’s reasoning are conditioned on the same demographic persona, researchers can hold environment and intervention constant while varying stratum — or vary the intervention while holding stratum constant — isolating the contribution of socioeconomic and life-stage factors to any behavioral outcome. The Tier 2 signal experiment (Table 8) illustrates this: ~~Type B educational signals drew equal or higher acceptance than direct commands for cost-sensitive strata (O2 and O4: 2 of 5 seeds each), while O1 (employed single adult) responded only to Type A direct commands (1 of 5) and O3 (employed parent) rejected all signal types~~ educational price signals (Type B) were accepted in every stratum, from one agent in five (O3) to three in five (O4), while direct commands were declined in all but one case and social comparison in every case — differences in direction that a fixed schedule cannot produce at all, having no mechanism to respond to a signal, ~~though with five seeds per cell they are indicative rather than precise.~~ The signal types correspond to established intervention typologies in behavioral energy

research [30, 33], ~~making results directly comparable to field studies so that~~
~~results can be set beside field studies that use the same typology.~~

600 **Platform-neutral integration.** The MCP server layer means that any tool
with MCP support—including Claude Desktop, LangChain, energy schedul-
ing systems [45], and custom orchestration frameworks—can drive the occupant
agent inside a building simulation ~~with demographically-differentiated-occupant~~
~~behavior~~—without writing integration code. The REST API provides an
605 equivalent path for EnergyPlus co-simulation and Home Assistant integra-
tions [12, 15].

Community extensibility. The plugin registry and abstract base classes
lower the barrier for researchers who want to add custom demographic strata,
alternative behavioral data sources or activity grounding pipelines, or novel
610 memory architectures. The conformance test suite enables rapid validation
of extensions before publication.

5. Conclusions

BUILD OCC delivers a survey-grounded, three-layer occupant agent platform
that the building energy community can use, extend, and build upon without
615 reconstructing the behavioral layer for each study. Tier 1 confirms that ~~ATUS~~
~~grounding produces empirically-calibrated activity distributions~~ the scheduler
reproduces the ATUS activity distributions to within sampling noise; Tier 2
shows that demographic priors propagate into differentiated agent reasoning,
though it does not by itself establish that the resulting behavior is realistic.
620 Beyond the implementation, the platform treats occupant behavior as a
structured, demographically parameterizable experimental variable rather
than an uncontrolled source of uncertainty. The ATUS-derived strata, frozen
schema, and plugin registry are intended to make that variable reproducible
and extendable across independent research groups.

625 Current limitations include a single-action constraint per timestep; ATUS-
grounded strata covering US demographic profiles only; activity grounding
restricted to primary activities, since ATUS records only one activity at a
time; and a memory-importance score that the agent assigns to itself and
then reuses for retrieval, without external calibration or a feedback path that
630 would correct a mis-scored entry. Subsequent phases extend the platform
with multi-agent household simulation, non-US occupancy data, and external
behavioral benchmarks including ecobee Smart Thermostat field data [46]
matched to the four ATUS strata.

CRediT Author Statement

635 **Wooyoung Jung:** Conceptualization, Methodology, Software, Validation,
Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares that there are no competing interests.

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Current executable software version

Table 9: Software metadata (mandatory)

Nr.	Software metadata de- scription	Please fill in this column
S1	Current software version	v1.0.1
S2	Permanent link to executables of this version	https://pypi.org/project/ buildocc/
S3	Legal software license	Apache License 2.0
S4	Computing platforms / OS	Linux, macOS, Windows (Python 3.11+)
S5	Installation requirements & dependencies	<code>pip install buildocc</code> ; an LLM provider API key (An- thropic, OpenAI, Google) or a local Ollama model
S6	Link to user manual	https://github.com/ humanbuildingsynergy/ Build0cc/blob/main/ README.md
S7	Support email for questions	wooyoungjung@arizona.edu

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A. ATUS Activity Probability Computation

This appendix provides the formal description of the activity probability computation summarized in Section 2.

Episode-overlap framing. ATUS diaries record each respondent's day as a sequence of episodes, each with a start time t_s (minutes since midnight), stop time t_e , and a six-digit activity code in ATUS. For ~~a 15-minute simulation slot centered at time τ , the scheduler identifies which activity each hour~~
820 h , the method identifies which episode was in progress at the ~~midpoint $\tau + 7.5$ min~~half-hour mark:

$$c(r, \tau h) = \text{activity code of respondent } r \text{ such that } t_s \leq \tau + 7.5 \text{ min } h:30 < t_e \quad (\text{A.1})$$

where r indexes individual diary respondents. This in-progress query avoids the start-time bias that would overrepresent short activities (~~e.g. under a start-time count, a 5-minute meal counts only if the midpoint falls within it~~)
825 and an 8-hour sleep would each count once).
ATUS diaries run from 04:00 to 04:00 the following day, and episode times are recorded as ordinary clock times. Episodes are therefore placed on a 04:00-anchored axis before the hourly query, so that an episode crossing midnight is credited to the diary day it belongs to.

830 *Category probability.* ~~Eight~~ Nine activity categories are defined (Table A.1). The probability of category k at hour h , for demographic stratum s and day type $d \in \{\text{weekday, weekend}\}$, is:

$$P(k \mid s, d, h) = \frac{\sum_{r \in R(s, d)} w_r \cdot \mathbf{1}[c(r, h:30) \in k]}{\sum_{r \in R(s, d)} w_r} \quad (\text{A.2})$$

where $R(s, d)$ is the set of diary respondents matching stratum s on day type d ; w_r is the ATUS person-level survey weight (provided by ~~BLS~~ the Bureau of Labor Statistics (BLS) to correct for sampling design); $\mathbf{1}[\cdot]$ is the indicator function that equals 1 when respondent r was engaged in a category- k activity at the half-hour mark $h:30$ of hour h ; and the denominator normalizes the weighted counts to probabilities. Evaluating Equation (A.2) for all $h \in \{0, \dots, 23\}$ and all ~~8~~ nine categories yields a ~~24 $\times$ 8~~ 24 $\times$ 9 weighted probability table per stratum per day type, which is bundled with the installed package.

Code-level sampling. Once a category k is sampled from $P(k \mid s, d, h)$, a specific activity code is drawn from within that category proportional to the total diary minutes logged across all stratum respondents for each code. This
845 gives common activities realistic relative frequencies (e.g., sleeping in a bed

Table A.1: The nine broad activity categories used in the two-step ATUS sampling procedure. ATUS activity codes are six-digit and hierarchical, so a filter is a leading prefix of the code; the third column counts the six-digit codes mapped to each category.

Category	ATUS code filter	Codes	Representative subcategories
Sleeping	prefix 0101	3	Sleeping; Sleeplessness
Work	prefix 05	14	Work (main job); Work (other job); Security procedures related to work
Food preparation	prefix 0202	4	Food and drink preparation; Food presentation; Kitchen and food clean-up
Laundry	code 020102 exactly	1	Laundry
Television	codes 120303, 120304	2	Television and movies; Television (religious)
Eating	prefix 1101	2	Eating and drinking
Exercise	prefix 13	42	Aerobics; Basketball; Biking; Swimming; Weightlifting
Travel	prefix 18	31	Travel related to work; to housework; to personal care
Other	residual	201	Grooming; Reading; Computer use; Social communication

is drawn far more often than napping on a sofa, even though both map to the “sleeping” category).

Work-from-home probability. Finally, the WFH probability p_{WFH} for each stratum is estimated from the ATUS TEWHERE variable — the “where were you when you did this activity?” field — restricted to employed respondents who worked on their diary day. p_{WFH} ~~equals the~~ is the share of the stratum’s survey-weighted fraction of those respondents who reported working at home for any work episode, computed separately for each stratum work minutes reported at home (the at-home column of Table C.1). A per-day Bernoulli draw from p_{WFH} then determines whether the simulated agent works from home on a given day.

B. Memory Retrieval and Reflection Mechanics

This appendix provides the mathematical detail for the memory retrieval and reflection logic implemented in `MemoryStream`, following the generative agent architecture of Park et al. [18].

Retrieval scoring. At each simulation timestep t , every stored memory entry m receives a combined retrieval score $s(m, t) \in [0, 1]$ that balances how recent the entry is against how important it was judged to be when written:

$$s(m, t) = 0.5 \tilde{r}(m, t) + 0.5 \tilde{i}(m) \quad (\text{B.1})$$

where $\tilde{r}(m, t)$ is the normalized recency score and $\tilde{i}(m)$ is the normalized importance score, both in $[0, 1]$ and weighted equally.

The recency component decays exponentially with a 24-hour half-life:

$$\tilde{r}(m, t) = 2^{-(t-t_m)/24\text{h}} \quad (\text{B.2})$$

where t_m is the simulation time (in hours) at which entry m was written, and $(t - t_m)$ is the elapsed time in hours since writing. The score equals 1.0 at the moment of creation and halves every 24 simulation hours, so memories from the previous day retain ~~roughly~~ half their recency weight while week-old entries contribute near zero.

The importance component normalizes the integer score $i(m)$ assigned by the LLM at write time:

$$\tilde{i}(m) = \frac{i(m)}{10} \quad (\text{B.3})$$

where $i(m) \in \{1, \dots, 10\}$ is an integer on a scale from 1 (routine, nothing unusual) to 10 (rare, extreme situation), self-assigned by the agent’s LLM in the same JSON response that produces the behavioral action. The top- k entries ranked by $s(m, t)$ are injected into the next prompt ($k = 5$ by default).

Reflection trigger. An importance accumulator A tracks the cumulative importance of all entries written since the last reflection. After each entry m is stored, $A \leftarrow A + i(m)$. When $A \geq \theta$ (default $\theta = 100$), reflection is triggered: a dedicated LLM call receives the 30 most recent entries by simulation time and returns exactly three synthesized behavioral insights (e.g., “this occupant pre-cools before peak hours to avoid peak tariffs”). Each insight is written back into the stream as a new entry with importance 9.0, and A resets to zero. At average importance 7–10 (high-salience events), reflection fires ~~roughly~~ every 10–14 timesteps (≈ 2.5 – 3.5 simulation hours); at average importance 2–3 (routine observations), every 33–50 timesteps (≈ 8 – 12 simulation hours). In the measured runs of Section 2, reflection fired 1.5 times per simulated day.

890 C. Demographic Strata Selection Rationale

The four strata were selected to maximize variation in the two behavioral dimensions most consequential for residential demand response research: *peak-hour occupancy schedule* and *household composition*. The selection is grounded in the ATUS 2022–2023 microdata that drives the platform itself [47, 48, 49]

895 .
 Table C.1 ~~summarises~~ summarizes the occupancy metrics and sample sizes derived from ATUS 2022–2023 ($n = 16,684$ respondents). O1 and O3 are absent from the home for the majority of ~~peak-grid~~ daytime hours (09:00–17:00), while O2 and O4 are home ~~throughout for most of~~ the day: ~~59.8%~~ 60.9% and ~~61.4%~~ 62.6% of O1 and O3 weekday hours in that window are spent in work activity, against 0.4% and ~~7.92%~~ 4% for O2 and O4. Within each occupancy cluster, household composition is varied to capture differences in thermostat authority and social dynamics: ~~single-person (O1, O4), two-adult couple (lives alone, O2), and family with children (is a couple, O3) has~~ children at home, and O4 carries no household restriction [30, 33]. O4 has the smallest sample (~~$n = 107$~~); ~~its 15-minute probability tables are the noisiest of the four strata~~ $n = 677$. Additional strata (multi-adult non-family, student dormitories) would draw from the remaining ~~10,643~~ 073 “other” respondents but would fragment already thin subsamples without a clearly defined ~~DR~~ demand response participation profile.

~~Current executable software version~~

Table C.1: ATUS 2022–2023 occupancy metrics and sample sizes by stratum. % working 09–17 is the mean weighted probability of being engaged in work activity during the 09:00–17:00 window on weekdays (Appendix A). Location columns split each stratum’s work time (ATUS activity codes 05, minutes weighted by TUFINLWGT, episodes with a valid location code) across the respondent’s home, a workplace, and elsewhere; the three sum to 100%. For O2 and O4 the split rests on the small amount of work time those strata report. The at-home share is the work-from-home probability of Section 2.

ID	Stratum	% working 09–17 (wkdy)	% at workplace	% at home	% else- where	n
O1	Employed single adult	60.9	70.2	24.3	5.6	1,393
O2	Retired couple	0.4	8.1	50.1	41.8	2,351
O3	Employed parent	62.6	68.9	25.7	5.4	2,190
O4	Not-employed adult	2.4	26.6	68.7	4.6	677



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07/04/2026

Dear Editor:

I am pleased to submit the manuscript "BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research" for consideration in SoftwareX.

BuildOcc is an open-source Python platform that grounds large language model (LLM) agents in the American Time Use Survey (ATUS) — a nationally representative diary dataset covering 16,684 respondents — to simulate building occupant behavior for energy research. Existing occupant behavior models rely on fixed schedules or simple stochastic processes that cannot reason about a person's history, current context, or the type of energy signal being delivered. Prior LLM-agent proposals in this space have used hand-crafted profiles or small convenience samples and have not released a general-purpose, survey-grounded platform for reuse. BuildOcc fills this gap. Each simulated occupant maintains a persona derived from ATUS population statistics, a timestep-level memory stream with periodic reflection, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface (Python library, REST API, and Model Context Protocol server) so that any building energy tool can integrate behavioral intelligence without bespoke coupling code. A plugin registry allows the community to add new demographic strata, custom schedulers, and alternative memory backends as separate installable packages. BuildOcc is an original, openly released software contribution (Apache License 2.0, v1.0.0 at github.com/humanbuildingsynergy/BuildOcc) that addresses a concrete gap in the building energy simulation community:

We believe that this article strongly contributes to the building energy research community. Also, it fits the scope and aim of SoftwareX and complies with its requirements. This manuscript has not been published and is not under consideration for publication elsewhere.

Thank you very much for your consideration.

Sincerely,

A handwritten signature in black ink, appearing to read 'WJung'.

Wooyoung Jung, Ph.D.

Assistant Professor, The University of Arizona

